Joint Estimation of Absolute Attitude and Size for Satellite Targets Based on Multi-Feature Fusion of Single ISAR Image

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***Abstract*— It is challenging to estimate satellite targets' absolute attitude and size with limited observational data. This article proposes an innovative way to jointly estimate satellite targets' absolute attitude and size in the 3-D stable coordinates based on inverse synthetic aperture radar (ISAR) image interpretation with only one image. By taking advantage of the rectangular solar panels commonly equipped on satellites, this article extracts solar panel's principal components, line features, and phase features of single ISAR imagery with principal component analysis (PCA), radon transform (RT), and minimum-entropy (ME)-based autofocus method, respectively. The projection relationship between these features and the absolute attitude and size of the satellite are established separately. Through multi-features fusion, a joint parameter estimation optimization function is established. This optimization is solved iteratively by the quasi-Newton method. The attitude and size parameters can be estimated simultaneously and rapidly, realizing the satellite state estimation under limited observation data. The excellent performance of the proposed algorithm is verified through different experiments.**

***Index Terms*— Attitude estimation, inverse synthetic aperture radar (ISAR) image interpretation, multi-features fusion, satellite targets, size estimation.**

I. Introduction

AS THE number of space targets operating in orbit continues to rise, the development and upgrading of space target surveillance systems have received significant attention from many countries [1]–[3]. The improvement of space target detection and identification capability has become an increasingly urgent requirement. The absolute attitude and size of space targets are important characteristic parameters reflecting the flight status of space targets. By estimating and analyzing the attitude of space targets, especially the important payloads of unknown non-cooperative space targets, we can obtain important information such as payload orientation, movement intention, and observation area [4]–[7]. Inverse synthetic aperture radar (ISAR) has the advantages of all-day, all-weather, long-range observation, and high-resolution imaging and plays an important role in executing non-cooperative satellite attitude measurement in space at a long distance [8]–[10]. The acquisition of space target attitude and size information based on high-resolution ISAR images can effectively determine target intention behavior and situation evolution and avoid potential space collisions, which is the focus of current space situational awareness research [6], [11]–[13].

Among the traditional satellite attitude estimation methods, they can usually be divided into two major categories. One category applies to cooperative satellites, which requires prior information of the target, including the satellite target's electromagnetic (EM) model. The method establishes a complete database through the full-angle observation of the model, combining with a large amount of observation data, such as radar cross section (RCS) [12], [14], [15], ISAR images [6], [7], [16], [17], optical images [18]–[21], and so on, and achieves the attitude estimation by extraction the features of these observation data and matching them with the template library. This method is suitable for cooperative satellites with EM models. Still, it will fail on unknown non-cooperative targets, and this method requires long-time observations of the target, making practical application difficult. Another category of the method is the 3-D reconstruction-based method [13], [22], [23], such as the factorization-based method, which can acquire the 3-D structure of the target based on the ISAR image sequences obtained from long-time observations, based on which the satellite's attitude and size can be estimated. Such methods require a large amount of observation data. They need the scatterer's association of the sequences of ISAR images, which is not an easy task in practice due to a large number of scatterers on the target. Moreover, there is an unknown rotation between the 3-D structure obtained by such methods and the actual construction of the target, and the absolute attitude estimation cannot be achieved.

Due to the shortcomings and practical inapplicability of the above traditional methods, some researchers have recently proposed the satellite attitude estimation method based on the projection principle of ISAR imaging. Since ISAR image is

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a projection of the target's 3-D structure to the 2-D imaging plane, such a direct mapping relationship allows one to estimate the satellite's 3-D absolute attitude from interpreting ISAR images, which provides a new idea for satellite attitude estimation. Zhou *et al.* [11], [24] and Wang *et al.* [25] innovatively proposed a method for satellite attitude estimation using the typical line structure of satellites. According to the serial observation view of the satellite, the method establishes the mapping relationship between the line features of multiple ISAR images and the 3-D attitude of the satellite, builds the optimization function, and solves it via particle swarm optimization (PSO) [26]. The advantage of this method is that the line features of satellites are commonly available. After all, almost every satellite is equipped with rectangular solar panels, which can be applied to non-cooperative targets without prior information. However, the method also requires long-time observations to obtain sufficient image sequences. The method estimates the satellite attitude and size separately, which does not take full advantage of the integral rectangular structure of the solar panels and will increase the estimated error.

Considering that the satellite attitude estimation methods based on long-time observations are difficult to apply in practical situations, Zhou *et al.* [27] proposed a satellite attitude estimation method based on the fusion of a single ISAR image and optical image in the literature. The method takes advantage of the difference and connection between the projection principles of optical and radar images under the same observation view. It combines the typical line characteristics of satellites to achieve attitude estimation based on the imaging projection principle. However, the method requires optical and radar sensors to work simultaneously in the same viewing angle, and the engineering implementation is costly. Moreover, in practice, the ground-based optical sensors suffer from low resolution in space target imaging, making the method difficult to practically apply. Zhou *et al.* [28] and [29] estimate the satellite's attitude using the quadratic spatial-variant phase error (QSVPE) generated by the target's rotation. The method requires only one image, which considers the satellite's solar panel as a plane. The QSVP coefficients generated by its rotation indirectly represent the norm vector of its solar panel. However, the method does not achieve the absolute attitude estimation because there is an unknown rotational angle between the absolute attitude and the estimated one around the normal vector. And the method cannot achieve the size estimation of the satellite.

Inspired by the literature [28], this article proposes an approach for joint estimation of absolute attitude and size for satellites based on the multi-feature fusion of a single ISAR image. Considering that literature [28] only utilizes a single satellite feature, that is, the QSVP feature and does not take full advantage of the satellite projection information in the ISAR image. This article will exploit different features of a single ISAR image, including the typical line features and point features of satellite projection, respectively. Combined with the QSVP feature in [28], the projection mapping relationship corresponding to the multi-features is established separately. By fusing these features, the parametric optimization function of the absolute attitude and size is established. Finally, the Broyden–Fletcher–Goldfarb–Shanno (BFGS)-based quasi-Newton iterative method [30] is employed to rapidly and accurately search the attitude and size parameters quickly and accurately, realizing the absolute attitude and size estimation of the satellite based on a single ISAR image under limited observation data. Compared with the existing methods, the proposed method has several advantages.

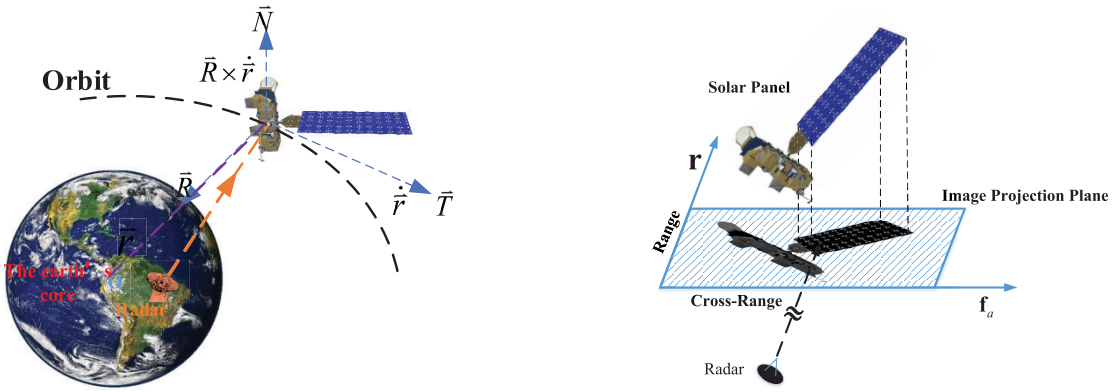
1) Theoretically, the proposed method requires only one ISAR image to achieve the joint estimation of satellite attitude and size. The proposed method is more suitable for practical applications than the methods that require long-time observations.
2) Compared with the method proposed in the literature [28], this article makes full use of the projection information of a single ISAR image via multi-feature fusion and can achieve both absolute attitude and absolute size estimation, instead of just the solar panel normal vector estimation in [28].
3) Compared with the PSO algorithm adopted in literature [11] and [28], this article employed a BFGS-based quasi-Newton iterative method, which dramatically reduces the algorithm's execution time and has a higher estimation accuracy than the PSO algorithm.

The remainder of the article is organized as follows. In Section II, the observation geometry and ISAR image projection model for the satellite targets are introduced. In Section III, we present the multi-features and the mapping relationships between each feature and the satellite attitude and size. Section IV introduces the extraction method of multi-features and established the parametric optimization function for joint estimation of the satellite's attitude and size. In Section V, a variety of experiments are designed, including experiments on satellite attitude and size estimation for single/multi-station observations at different poses. In Section VI, some conclusions are drawn.

II. Observation Geometry and Imaging Projection Principle

In this section, we will present the radar observation geometry model of an in-orbit satellite. In the satellite's stabilization coordinate system, the satellite is considered an attitude stabilization target under the attitude controller [31]. As shown in Fig. 1(b), the satellite is assumed to be observed in the local orbital frame X-Y-Z, which is called RTN coordinate [12], RTN coordinate system is defined by the radial (R), along-track (T), and cross-track (N) directions. This is a typical coordinate system that provides an essential reference for describing the attitude of an earth pointing satellite [3], [5]. Under this assumption, the relative rotational motion between the satellite and radar can be simply described by the variation of the radar line of sight (LOS) in the RTN coordinate. In this way, the ISAR observation geometry is described with the classical turn-table model [28], as shown in Fig. 1(b). In Fig. 1(b), the LOS is defined by the observation angles, that is, the azimuth angle $\alpha(t_m)$ and the elevation angle $\beta(t_m)$, where t_m is the slow time. The time-varying LOS $\mathbf{i}(t_m)$ can

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(a) (b)

Fig. 1. (a) Radar observation geometry of satellite in orbit. (b) ISAR imaging projection plane.

then be represented as

$$\mathbf{i}(t_m) = [\cos\beta(t_m)\sin\alpha(t_m), \cos\beta(t_m)\cos\alpha(t_m), \sin\beta(t_m)]^T. \tag{1}$$

In the satellite's stabilization coordinate system, the observation angles $\alpha(t_m)$ and $\beta(t_m)$ can be obtained from the Kepler two-body motion property [31], and the radar geographic coordinates, with the satellite orbital roots, are known. Zhou *et al.* [11] provide three methods for solving the radar observation angles and the second method is referred to in this article. The ISAR signal is expressed by the superposition of isolated point scatterer responses. The linear-frequency-modulated (LFM) signals are adopted in the imaging radar, that is,

$$s(\hat{t};t_m) = \mathrm{rect}\left(\frac{\hat{t}}{T_P}\right)\cdot\exp\left[j2\pi\left(f_c t + \frac{1}{2}\gamma\hat{t}^2\right)\right] \tag{2}$$

where

$$\mathrm{rect}(u) = \begin{cases} 1, & |u| \le \dfrac{1}{2} \\ 0, & |u| > \dfrac{1}{2} \end{cases}$$

T_P, f_c, and γ denotes the pulsewidth, carrier frequency, and frequency modulation rate. $t = \hat{t}+t_m$ is the full time, where $\hat{t}$ is the fast time, t_m is the slow time. Suppose that the distance between the scatter $P(x_R, y_T, z_N)$ and imaging radar at the time t_m is $R_p(t_m)$, then, the radar echo of this scatter can be written as

$$s(\hat{t};t_m) = \sigma_p \mathrm{rect}\left(\frac{\hat{t}-2R_p(t_m)/c}{T_P}\right)\cdot\exp\left[j2\pi\left(f_c(t-2R_p(t_m)/c) + \frac{1}{2}\gamma\left(\hat{t}-2R_p(t_m)/c\right)^2\right)\right] \tag{3}$$

where c and σ_p represent the speed of light and the reflection coefficient, respectively. Assume that the conventional range compression and range cell migration correction are well performed, (3) can be expressed as

$$s_p(\hat{t};t_m) = \sigma_p\cdot\mathrm{sinc}\left[B\cdot\left(\hat{t}-\frac{2\cdot R'_p(t_m)}{c}\right)\right]\cdot\exp\left[-j\frac{4\pi\cdot R'_p(t_m)}{\lambda}\right]. \tag{4}$$

In the turn-table model, $R'_p(t_m)$ is the variation of range caused by the time-varying LOS $\mathbf{i}(t_m)$ and $R'_p(t_m)$ can be expressed as [28]

$$R'_p(t_m) = \mathbf{P}\cdot\mathbf{i}(t_m) \approx R_0 + R_1 t_m + R_2 t_m^2 \tag{5}$$

where

$$\begin{cases} R_0 = \mathbf{i}^T(t_0)\mathbf{P} \\ R_1 = \mathbf{i}'^T(t_0)\mathbf{P} \\ R_2 = \mathbf{i}''^T(t_0)\mathbf{P} \end{cases} \tag{6}$$

where $\mathbf{i}^T(t_0)$, $\mathbf{i}'^T(t_0)$, and $\mathbf{i}''^T(t_0)$ are as, (7)–(9), shown at the bottom of the page. In (5), the higher-order terms of

$$\mathbf{i}(t_0) = [\cos\beta(t_0)\sin\alpha(t_0), \cos\beta(t_0)\cos\alpha(t_0), \sin\beta(t_0)]^T \tag{7}$$

$$\mathbf{i}'(t_0) = \begin{bmatrix} \cos\beta(t_0)\cos\alpha(t_0)\cdot\dot{\alpha} - \sin\beta(t_0)\sin\alpha(t_0)\cdot\dot{\beta} \\ -\cos\beta(t_0)\sin\alpha(t_0)\cdot\dot{\alpha} - \sin\beta(t_0)\cos\alpha(t_0)\cdot\dot{\beta} \\ \cos\beta(t_0)\cdot\dot{\beta} \end{bmatrix} \tag{8}$$

$$\mathbf{i}''(t_0) = \begin{bmatrix} \cos\beta(t_0)\cos\alpha(t_0)\cdot\ddot{\alpha} - \sin\beta(t_0)\sin\alpha(t_0)\cdot\ddot{\beta} - \sin\beta(t_0)\cos\alpha(t_0)\cdot\dot{\alpha}\dot{\beta} \\ -\cos\beta(t_0)\sin\alpha(t_0)\cdot\ddot{\alpha} - \sin\beta(t_0)\cos\alpha(t_0)\cdot\ddot{\beta} + \sin\beta(t_0)\sin\alpha(t_0)\cdot\dot{\alpha}\dot{\beta} \\ \cos\beta(t_0)\cdot\ddot{\beta} \end{bmatrix} \tag{9}$$

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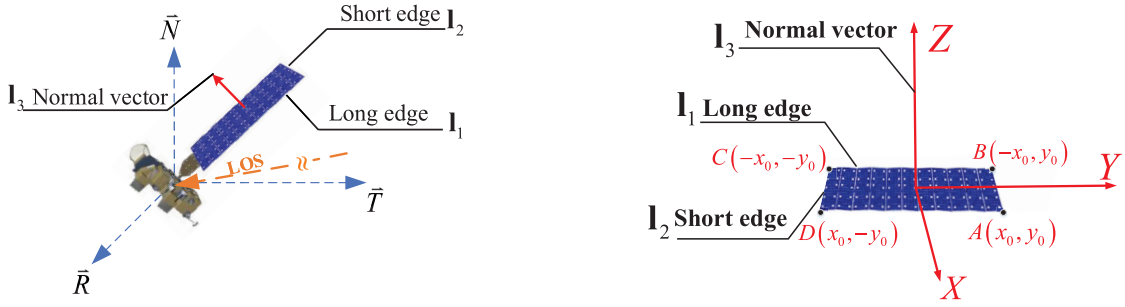


Fig. 2. (a) RTN coordinate system. (b) Defined solar panel coordinate system.

distance variation are ignored due to the small impact on ISAR imaging. In ISAR imaging, the distance variation due to the rotation is considered to be no more than one range gate. The quadratic phase error, which can be used to estimate the solar panel normal vector during the imaging auto-focus process (which will be described in detail in Section III), is assumed to be ideally compensated. By performing fast Fourier transform along azimuth direction, (4) can be expressed as

$$s_p(\hat{t};f_a) \approx \sigma_p \text{sinc}\left[B\left(\hat{t}-\frac{2R_0}{c}\right)\right]\sin c\left[T_a\left(f_a-\frac{2R_1}{\lambda}\right)\right]. \tag{10}$$

Equation (10) is the final ISAR image. From (10) as well as (4), (7), and (8), it can be seen that the projection relationship between the ISAR image and the target 3-D structure can be expressed as

$$\hat{\mathbf{P}} = \mathbf{M}\mathbf{P} \tag{11}$$

where $\hat{\mathbf{P}} = [u_p, v_p]^T$ is the coordinate of a point P on the image plane, $\mathbf{M} = \begin{bmatrix} \mathbf{i}^T(t_0)/\Delta r \\ \mathbf{i}'^T(t_0)/\Delta f_a \end{bmatrix}$ the projection matrix, $\Delta r = c/2B$, $\Delta f_a = \lambda/2\theta$ the range resolution and cross-range resolution, respectively.

The above is the projection principle of ISAR imaging, where the projection matrix $\mathbf{M}$ can be obtained by the three methods mentioned in [28]. The projection matrix $\mathbf{M}$ links the actual 3-D structure of the satellite target with its 2-D structure in the ISAR image, making it possible to estimate the absolute attitude and size of the satellite from the ISAR image. In addition, according to the method proposed in [28], the quadratic phase error in (5) is considered useful information that can be used to estimate the normal vector of the solar panel. In Section IV, we will describe how to establish the mapping relations between the multi-feature and the absolute attitude and size of the satellite using the projection principle of imaging and the quadratic phase features, from which the parametric optimization function of the absolute attitude and size are established.

III. Methodology Description and Multi-Feature Description

In general, for an attitude-stabilized satellite, its attitude is fixed in the RTN coordinate system. Its attitude can be represented by three mutually orthogonal vectors, the long edge, the short edge, and the normal vector of the solar panel, as shown in Fig. 2(a). Therefore, the estimation of the absolute attitude of the satellite can be converted into the estimation of the long edge vector $\mathbf{l}_1$, short edge vector $\mathbf{l}_2$, and normal vector $\mathbf{l}_3$ of the satellite solar panel. Considering the peculiarities of the satellite solar panel geometry (generally rectangular panels), we established a new coordinate system O-XYZ on the solar panel called the solar panel coordinate system. The three axes of its coordinate system are parallel to the long edge, short edge, and normal vector, respectively, as shown in Fig. 2(b). Next, we describe how to define its attitude parameters and size parameters in the RTN coordinate system.

In fact, the O-XYZ coordinate system can be transformed into the RTN coordinate system after certain rotational operations. Assume that the coordinates of an arbitrary point in the O-XYZ coordinate system is $\mathbf{Q}_i = [x_i, y_i, z_i]^T$, then its coordinates in the RTN coordinate system is

$$\mathbf{P}_i = \mathbf{R}(\theta,\varphi,\gamma)\mathbf{Q}_i \tag{12}$$

where $\mathbf{R}(\theta,\varphi,\gamma)$ is the rotation matrix between the O-XYZ coordinate system to the RTN coordinate system, donated as

$$\mathbf{R}(\theta,\varphi,\gamma) = \mathbf{R}_z(\theta)\cdot\mathbf{R}_y(\varphi)\cdot\mathbf{R}_x(\gamma) \tag{13}$$

where θ, φ, and γ are the rotational angles around Z-, Y-, and X-axis, respectively. $\mathbf{R}_z(\theta)$, $\mathbf{R}_y(\varphi)$, and $\mathbf{R}_x(\gamma)$ can be expressed as follows:

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{14}$$

$$\mathbf{R}_y(\varphi) = \begin{bmatrix} \cos\varphi & 0 & -\sin\varphi \\ 0 & 1 & 0 \\ \sin\varphi & 0 & \cos\varphi \end{bmatrix} \tag{15}$$

$$\mathbf{R}_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\gamma & -\sin\gamma \\ 0 & \sin\gamma & \cos\gamma \end{bmatrix}. \tag{16}$$

Noting (11) and (12), the mapping relationship between the O-XYZ coordinate system and the ISAR imaging plane can be expressed as

$$\hat{\mathbf{P}}_i = \mathbf{M}\mathbf{R}(\theta,\varphi,\gamma)\mathbf{Q}_i = \hat{\mathbf{M}}\mathbf{Q}_i. \tag{17}$$

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In the O-XYZ coordinate system, the unit vectors of the long edge, short edge, as well as the normal vector can be represented by

$$\begin{cases} \mathbf{n}_l = [1,0,0]^T \\ \mathbf{n}_s = [0,1,0]^T \\ \mathbf{n}_n = [0,0,1]^T. \end{cases} \tag{18}$$

Combined with (12), they can be expressed in the RTN coordinate system as $\mathbf{l}_1 = \mathbf{R}(\theta,\varphi,\gamma)\mathbf{n}_l$, $\mathbf{l}_2 = \mathbf{R}(\theta,\varphi,\gamma)\mathbf{n}_s$ and $\mathbf{l}_3 = \mathbf{R}(\theta,\varphi,\gamma)\mathbf{n}_n$. Here, $\mathbf{l}_1$, $\mathbf{l}_2$ and $\mathbf{l}_3$ are a sufficient representation of the absolute attitude of the satellite.

We will present the relationship between the multi-feature and the satellite's absolute attitude and size in the following.

A. Normal Vector of Solar Panel

Zhou *et al.* [28] exploited the solar panel's plane structure and presented the relationship between the QSVPE and the solar panel's normal vector. In the RTN coordinate system, scattering points distributed in a 2-D plane $P(x_R, y_T, z_N)$ can be expressed as a set with the inner product expression

$$\mathbf{l}_3 \cdot \mathbf{P}(x_R, y_T, z_N) = \varepsilon \tag{19}$$

where $\mathbf{l}_3$ represents the normal vector of the plane component and ε is the distance of the origin from the plane and equals 0 when the coordinate origin is contained in the plane. On the other hand, the QSVP is proportional to the range and Doppler coordinate of the scattering point in the RD image. It is expressed as a linear function in the relevant literature [30], [32], [33]

$$R_2 = e_r R_0 + e_a R_1 \tag{20}$$

where e_r and e_a are the QSVPE coefficients, which will be obtained in Section IV by minimum entropy (ME) optimization, and here we assume they are known. Bringing 6 into 20, an explicit expression bridging the defocusing performance and the normal vector is obtained

$$e_r \mathbf{i}^T(t_0)\mathbf{P} + e_a \mathbf{i}'^T(t_0)\mathbf{P} - \mathbf{i}''^T(t_0)\mathbf{P} = \mathbf{l}_3 \cdot \mathbf{P}(x_R, y_T, z_N) = \varepsilon. \tag{21}$$

This means that $\hat{\mathbf{l}}_3 = e_r \mathbf{i}^T(t_0) + e_a \mathbf{i}'^T(t_0) - \mathbf{i}''^T(t_0)$, where $\hat{\mathbf{l}}_3$ is the estimated normal vector of solar panel [28]. At this point, the error function between the estimated normal vector and the actual normal vector can be established and expressed as follows:

$$J_1(\theta,\varphi,\gamma) = \arg\min_{\theta,\varphi,\gamma} \left\{ \arccos\left[\mathbf{l}_3^T(\theta,\varphi,\gamma) \cdot \hat{\mathbf{l}}_3 \right] \right\}. \tag{22}$$

There are three unknown parameters in (22), θ, φ, and γ represent the 3-D rotation of the satellite from the solar panel coordinate system to the RTN coordinate system, that is, the absolute attitude of the satellite. However, there is only one equation. The attitude parameters cannot be solved. In literature [28], the estimated normal vector is directly treated as the satellite's attitude, which is not precisely accurate, as seen from the above analysis. In Sections III-B and III-C, other features will be associated with meet the solution.

B. PCA Process for the Solar Panel

Equation (17) gives the projection relationship between the solar panel coordinate system, that is, the O-XYZ coordinate system, to the ISAR imaging plane, and the projection matrix $\hat{\mathbf{M}}$ contains the absolute satellite attitude parameters θ, φ, and γ. Assume that the four vertices of the solar panel are A, B, C, D, located in the four quadrants of the X-O-Y plane, where the coordinate of the vertex A is $[x_0, y_0, 0]^T$. Then the coordinates of vertexes B, C, and D are $[-x_0, y_0, 0]^T$, $[-x_0, -y_0, 0]^T$, $[x_0, -y_0, 0]^T$. The solar panel's length and width are $2x_0$ and $2y_0$. The orientations of long and short edges are $\mathbf{n}_l = [1,0,0]^T$ and $\mathbf{n}_s = [0,1,0]^T$, respectively. Set $\mathbf{Q}$ as the location set of the points on the solar panel, which can be expressed as follows:

$$\mathbf{Q} = \begin{bmatrix} x_1 & x_2 & \cdots & x_K \\ y_1 & y_2 & \cdots & y_K \\ z_1 & z_2 & \cdots & z_K \end{bmatrix} \tag{23}$$

where K is the number of points on the solar panel. In the O-XYZ coordinate system, since the solar panel can be considered as a plane, so $y_i = 0$, $i = 1, 2, \ldots, K$. For a set of points uniformly distributed in a rectangular region, its scattering point coordinates should satisfy a uniform distribution, that is, as shown in the following:

$$\begin{cases} x \sim U(-x_0, x_0) \\ y \sim U(-y_0, y_0). \end{cases} \tag{24}$$

The mean and variance of the points on the solar panel are

$$\begin{cases} \bar{x} = 0, \quad \bar{y} = 0 \\ D_x = \dfrac{x_0^2}{3}, \quad D_y = \dfrac{y_0^2}{3}. \end{cases} \tag{25}$$

The PCA process is solving the eigenvalues and eigenvectors of the datasets covariance [25]. Here, the covariance of the dataset $\mathbf{Q}$ is expressed as

$$\begin{aligned} \mathbf{C}_\mathbf{Q} &= \frac{1}{K}\mathbf{Q}\mathbf{Q}^T \\ &= \frac{1}{K}\begin{bmatrix} \sum_{i=1}^{K} x_i^2 & \sum_{i=1}^{K} x_i y_i & \sum_{i=1}^{K} x_i z_i \\ \sum_{i=1}^{K} x_i y_i & \sum_{i=1}^{K} y_i^2 & \sum_{i=1}^{K} y_i z_i \\ \sum_{i=1}^{K} x_i z_i & \sum_{i=1}^{K} y_i z_i & \sum_{i=1}^{K} z_i^2 \end{bmatrix}. \end{aligned} \tag{26}$$

Since x, y, and z are independent of each other and $z_i = 0$, (26) can be simplified as

$$\mathbf{C}_\mathbf{Q} = \begin{bmatrix} D_x & 0 & 0 \\ 0 & D_y & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{E}\Lambda\mathbf{E}^T \tag{27}$$

where

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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is a matrix of eigenvectors and

$$\Lambda = \begin{bmatrix} D_x & 0 & 0 \\ 0 & D_y & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is a diagonal matrix containing eigenvalues. Notice that the eigenvectors composed by $\mathbf{E}$ are the long and short edges' orientations of the solar panel, while Λ containing eigenvalues that correspond to the lengths of the long and short edges of the solar panel.

Here, we assume that the set of solar panel points on the ISAR image is $\hat{\mathbf{P}} = \begin{bmatrix} u_1 & u_2 & \cdots & u_K \\ v_1 & v_2 & \cdots & v_K \end{bmatrix}$, then its covariance matrix is

$$\begin{aligned} \mathbf{C}_{\hat{\mathbf{P}}} &= \frac{1}{K}\hat{\mathbf{P}} \cdot \hat{\mathbf{P}}^T = \frac{1}{K}\begin{bmatrix} \sum_{i=1}^{K} u_i^2 & \sum_{i=1}^{K} u_i v_i \\ \sum_{i=1}^{K} u_i v_i & \sum_{i=1}^{K} v_i^2 \end{bmatrix} \\ &= \begin{bmatrix} D_u & D_{uv} \\ D_{vu} & D_v \end{bmatrix} \end{aligned} \tag{28}$$

where (u_i, v_i) is the coordinate of the ith scatterer in ISAR image, D_u and D_v are the variance of range and cross-range coordinates, respectively. D_{uv} is their covariance. On the other hand, since $\hat{\mathbf{P}} = \hat{\mathbf{M}}\mathbf{Q}$, $\mathbf{C}_{\hat{\mathbf{P}}}$ can be expressed as

$$\mathbf{C}_{\hat{\mathbf{P}}} = \frac{1}{K}\hat{\mathbf{P}} \cdot \hat{\mathbf{P}}^T = \frac{1}{K}\hat{\mathbf{M}}\mathbf{Q}\mathbf{Q}^T\hat{\mathbf{M}}^T = \hat{\mathbf{M}}\mathbf{C}_{\mathbf{Q}}\hat{\mathbf{M}}^T. \tag{29}$$

Equation (29) establishes the connection between $\mathbf{C}_{\mathbf{Q}}$ and $\mathbf{C}_{\hat{\mathbf{P}}}$. With covariance matrix $\mathbf{C}_{\hat{\mathbf{P}}}$, we can get its eigenvectors $\mathbf{V}_1$ and $\mathbf{V}_2$, and the eigenvalues λ_1 and λ_2. Detailed expressions for $\mathbf{C}_{\hat{\mathbf{P}}}$, $\mathbf{V}_1$, $\mathbf{V}_2$, λ_1, and λ_2 are given in Appendix A. Notice (68), where D_u, D_v, and D_{uv} are functions of the variables $(\theta, \varphi, \gamma, x_0, y_0)$, which makes the eigenvectors and eigenvalues in (69) and (70) also functions of those variables. Assume that the extracted eigenvalues and eigenvectors of the point set on ISAR image are $\tilde{\lambda}_1$, $\tilde{\lambda}_2$, $\tilde{\mathbf{V}}_1$, and $\tilde{\mathbf{V}}_2$, and their true values are $\lambda_1(\Theta)$, $\lambda_2(\Theta)$, $\mathbf{V}_1(\Theta)$, and $\mathbf{V}_2(\Theta)$, where $\Theta = [\theta, \varphi, \gamma, x_0, y_0]^T$ are the attitude and size parameters. Based on the above analysis, the fitness function can be established as

$$J_2(\Theta) = \arg\min_{\Theta} \left\{ \begin{array}{l} \sum_{i=1}^{2} \left| \lambda_i(\Theta) - \tilde{\lambda}_i \right| \\ \quad + \sum_{j=1}^{2} \arccos\left[\mathbf{V}_j^T(\Theta) \cdot \tilde{\mathbf{V}}_j \right] \end{array} \right\}. \tag{30}$$

Combining (22) and (30), it can be seen that five unknown parameters correspond to five equations. Theoretically, the absolute attitude and size parameters $\Theta = [\theta, \varphi, \gamma, x_0, y_0]^T$ can be solved by associating (22) and (30). However, to ensure the robustness of the proposed method, we need more features to ensure the stability of the proposed method in practical applications. The mapping relationship between the line features and the attitude and size parameters will be presented below.

C. Line Features of Solar Panel

With the previous analysis, it is known that in the RTN coordinate system, the long and short edges of the solar panel can be expressed as

$$\begin{cases} \mathbf{l}_1 = \mathbf{R}(\theta, \varphi, \gamma)\mathbf{n}_l \\ \mathbf{l}_2 = \mathbf{R}(\theta, \varphi, \gamma)\mathbf{n}_s. \end{cases} \tag{31}$$

Combining (17), the projection vectors of the long and short edges in the ISAR image can be expressed as

$$\begin{cases} \hat{\mathbf{l}}_1(\theta, \varphi, \gamma) = \mathbf{M}\mathbf{l}_1 = \hat{\mathbf{M}}\mathbf{n}_l \\ \hat{\mathbf{l}}_2(\theta, \varphi, \gamma) = \mathbf{M}\mathbf{l}_2 = \hat{\mathbf{M}}\mathbf{n}_s. \end{cases} \tag{32}$$

Assuming that the long and short edge vectors extracted from the ISAR image are $\tilde{\mathbf{l}}_1$ and $\tilde{\mathbf{l}}_2$, and the projection lengths are L_1 and L_2, the fitness function can be established as follows:

$$J_3(\Theta) = \arg\min_{\Theta} \left\{ \begin{array}{l} \left\| 2x_0\hat{\mathbf{l}}_1(\theta, \varphi, \gamma) - L_1\tilde{\mathbf{l}}_1 \right\|_2 \\ \quad + \left\| 2y_0\hat{\mathbf{l}}_2(\theta, \varphi, \gamma) - L_2\tilde{\mathbf{l}}_2 \right\|_2 \end{array} \right\} \tag{33}$$

where $\| \cdot \|_2$ denotes the L2-norm.

The above are the mapping relationships between multi-features and absolute attitude and size parameters [11], [24]. Based on (22), (30), and (33), the joint estimation optimization function for satellite attitude and size can be established as

$$J(\Theta) = J_1(\theta, \varphi, \gamma) + J_2(\Theta) + J_3(\Theta). \tag{34}$$

In Section IV, we will present how to extract multi-feature using different ISAR image interpretation methods. In addition, a quasi-Newton approach is adopted to search the attitude and size parameters in a fast and accurate way.

IV. Multi-Feature Extraction and Parametric Optimization

A. Solar Panel Segmentation

As we have analyzed in Section III, this article exploits the rectangular structure characteristic of the satellite solar panel. First, we need to segment the solar panel from the ISAR image. Zhou *et al.* [11] used a simple way to extract the solar panel region, that is, to determine whether the image pixels are within the four vertex regions of the solar panel. This is a rough way of segmentation, and since we need to extract the set of points on the solar panel, a more precise way is necessary. In our previous work, we proposed a satellite solar panel segmentation method based on the Pix2pixGAN network, refer to [25]. In addition, in some tasks, we can also use manual extraction to segment solar panels. After all, the proposed method does not require much data. Since this is not the focus of this article, the segmentation process will not be discussed here, and the specific methods can be found in literature [11] and [25].

B. Normal Vector Estimation Based on MEA

From (21), it can be seen that to obtain the normal vector $\hat{\mathbf{l}}_3$ of the solar panel, the QSVP coefficients e_r and e_a must be obtained. From (20), it can be seen that the QSVP can be expressed as a linear combination of the constant phase R_0

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and the primary phase R_1. Considering (4), the signal with QSVPE in the range time and azimuth time domain can be expressed as

$$\begin{aligned} s_p(\hat{t};t_m) &\approx \sigma_p \sin c\left[B\cdot\left(\hat{t}-\frac{2R_0}{c}\right)\right] \\ &\quad\cdot\exp\left[-j4\pi\frac{R_0+R_1t_m+R_2t_m^2}{\lambda}\right] \\ &\approx \tilde{s}_p(\hat{t};t_m)\cdot\exp\left[-j4\pi\frac{e_rR_0+e_aR_1}{\lambda}t_m^2\right] \end{aligned} \tag{35}$$

where $\tilde{s}_p(\hat{t};t_m)$ is the ideal image without phase error, expressed as follows:

$$\tilde{s}_p(\hat{t};t_m)=\sigma_p \sin c\left[B\cdot\left(\hat{t}-\frac{2R_0}{c}\right)\right]\cdot\exp\left[-j4\pi\frac{R_1t_m}{\lambda}\right] \tag{36}$$

where the constant term $\exp[-j4\pi(R_0/\lambda)]$ is ignored, considering that R_0 and R_1 can be expressed as range and cross-range coordinates, that is, $R_0=(c/2B)k$ and $R_1=(\lambda/2T_a)h$, where k is the range index, h is the azimuth aperture index, and T_a is coherent accumulation time. So, (35) can be expressed in discrete form as [30]

$$s_p(k;n)\approx\tilde{s}_p(k;n)\cdot\exp\left[-j(ak+bh)n^2\right] \tag{37}$$

where $a=((2\pi ce_r)/\lambda B)$, $b=(2\pi e_a/T_a)$, and n is the discrete form of slow time t_m. The azimuth phase error compensation for (37) gives

$$\tilde{s}_p(k;h)\approx s_p(k;h)\cdot\exp\left[j(ak+bh)n^2\right]. \tag{38}$$

The focused image is obtained by fast Fourier transforming (FFT) along the azimuth direction of (38)

$$\begin{aligned} \tilde{g}(k,h;a,b)=\frac{1}{N}\sum_{n=0}^{N-1}\exp\left(j2\pi\frac{hn}{N}\right) \\ \cdot s_p(k;h)\cdot\exp\left[j(ak+bh)n^2\right]. \end{aligned} \tag{39}$$

Entropy is one of the most primary functions in error compensation for ISAR imaging, and it has been used as a typical indicator in ISAR imaging in many different ways [34]–[36]. According to (39), the entropy of an image is defined as a function of (a,b), which is given by

$$\begin{aligned} E_g(a,b)=\ln S_g-\frac{1}{S_g}\sum_{k=0}^{K-1}\sum_{h=0}^{N-1}|\tilde{g}(k,h;a,b)|^2 \\ \cdot\ln|\tilde{g}(k,h;a,b)|^2 \end{aligned} \tag{40}$$

where S_g is the image intensity that can be expressed as

$$S_g=\sum_{k=0}^{K-1}\sum_{h=0}^{N-1}|\tilde{g}(k,h;a,b)|^2. \tag{41}$$

Then the estimation of (a,b) is obtained by minimizing the image entropy, expressed as follows:

$$(\hat{a},\hat{b})=\arg\min_{a,b}E_g(a,b). \tag{42}$$

According to the introduction in literature [30], the optimization of (42) can be solved by most quasi-Newton algorithms, such as the BFGS algorithm. First, we need to derive the partial derivatives of a and b. Fig. 3 gives the variation of

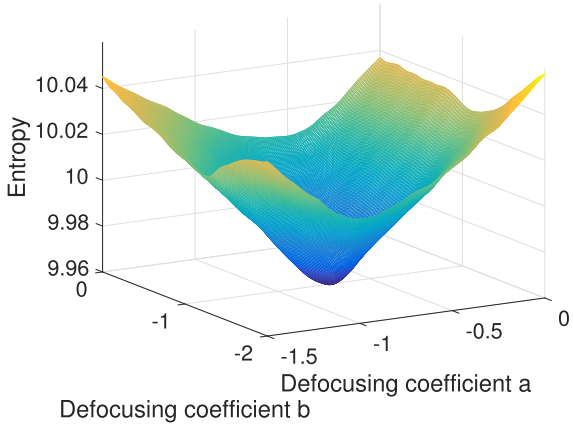


Fig. 3. Variation of image entropy with defocusing coefficients a and b.

satellite ISAR image entropy with the defocusing coefficients a and b, and it can be seen that the image entropy is minimized when the defocusing coefficients a and b are obtained optimally

$$\begin{aligned} &\frac{\partial E_g(a,b)}{\partial a} \\ &=-\frac{1}{S_g}\sum_{k=0}^{K-1}\sum_{h=0}^{N-1}\left[1+\ln|\tilde{g}(k,h;a,b)|^2\right]\frac{\partial|\tilde{g}(k,h;a,b)|^2}{\partial a} \\ &=-\frac{1}{S_g}\sum_{k=0}^{K-1}\sum_{h=0}^{N-1}\left[1+\ln|\tilde{g}(k,h;a,b)|^2\right] \\ &\quad\cdot\left\{2\mathrm{Re}\left[\tilde{g}^*(k,h;a,b)\frac{\partial\tilde{g}(k,h;a,b)}{\partial a}\right]\right\} \end{aligned} \tag{43}$$

where

$$\begin{aligned} \frac{\partial\tilde{g}(k,h;a,b)}{\partial a}=\frac{1}{N}\sum_{n=0}^{N-1}\exp\left(j2\pi\frac{hn}{N}\right) \\ \cdot s_p(k;h)\cdot\exp\left[j(ak+bh)n^2\right]\cdot\left(jkn^2\right). \end{aligned} \tag{44}$$

Similar to the partial derivatives of a, the partial derivatives of b can be expressed as

$$\begin{aligned} \frac{\partial\tilde{g}(k,h;a,b)}{\partial b}=\frac{1}{N}\sum_{n=0}^{N-1}\exp\left(j2\pi\frac{hn}{N}\right) \\ \cdot s_p(k;h)\cdot\exp\left[j(ak+bh)n^2\right]\cdot\left(jhn^2\right). \end{aligned} \tag{45}$$

With the partial derivative expressions (44) and (45), the gradient of image entropy with respect to (a,b) is

$$\nabla E_g(a,b)=\left[\frac{\partial E_g(a,b)}{\partial a},\frac{\partial E_g(a,b)}{\partial b}\right]^T. \tag{46}$$

Let $\mathbf{x}_0=[0,0]^T$ be the initial parameter vector $\mathbf{x}_k$ be the parameter vector of the kth iteration. The searching direction in BFGS is updated as follows:

$$\mathbf{d}_k=-\mathbf{B}_k\cdot\nabla E_g(a,b). \tag{47}$$

Then the $k+1$th parameter vector can be expressed as

$$\mathbf{x}_{k+1}=\mathbf{x}_k+\rho_k\mathbf{d}_k \tag{48}$$

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where ρ_k is the searching step that can be estimated by (49) via some 1-D inexact searching methods, such as golden section search or the Armijo–Goldstein step size rule

$$\rho_k = \arg\min_{\rho_k} \left(E_g(\mathbf{x}_k + \rho_k \mathbf{d}_k) \right). \tag{49}$$

The Hessian matrix $\mathbf{B}_k$ in BFGS is updated as follows:

$$\mathbf{B}_{k+1} = \mathbf{B}_k + \frac{\mathbf{y}_k \cdot \mathbf{y}_k^T}{\mathbf{y}_k \cdot \mathbf{s}_k^T} - \frac{\mathbf{B}_k \mathbf{s}_k \mathbf{s}_k^T \mathbf{B}_k}{\mathbf{s}_k^T \mathbf{B}_k \mathbf{s}_k} \tag{50}$$

where $\mathbf{s}_k = \rho_k \mathbf{d}_k$ and $\mathbf{y}_k = \nabla E_g^{k+1}(a,b) - \nabla E_g^k(a,b)$. The iteration can be terminated when $\|\nabla E_g^k(a,b)\| \le \varepsilon$ or $E_g^{k+1}(a,b) - E_g^k(a,b) \le \varepsilon$, where ε is the iteration stop threshold, typically $\varepsilon = 10^{-3}$. When the iteration ends, the optimal parameter vector $\mathbf{x}_{\text{opt}} = [\hat{a}, \hat{b}]^T$ is determined. The QSVP coefficients e_r and e_a can be obtained as follows:

$$\begin{cases} \hat{e}_r = \dfrac{\hat{a}\lambda B}{2\pi c}, \quad \hat{e}_a = \dfrac{\hat{b} T_a}{2\pi}. \end{cases} \tag{51}$$

Then the normal vector of solar panel can be obtained as

$$\hat{\mathbf{I}}_3 = \hat{e}_r \mathbf{i}^T(t_0) + \hat{e}_a \mathbf{i}'^T(t_0) - \mathbf{i}''^T(t_0). \tag{52}$$

It is worth mentioning that this section compensates the QSVPE while obtaining the coefficients e_r and e_a, which brings a well-focused image and guarantees the following extraction of point and line features.

C. Principal Components Extraction of the Solar Panel

To obtain the principal components of the solar panel of the ISAR image, that is, the eigenvalues $\tilde{\lambda}_1$ and $\tilde{\lambda}_2$, the eigenvectors $\tilde{\mathbf{V}}_1$ and $\tilde{\mathbf{V}}_2$, the point set of the solar panel should be extracted first. To ensure computational efficiency, the peak extraction technique is adopted in this article. In [37], the peak extraction technique combined with the CLEAN technique can be summarized as follows: in each iteration, search the position and amplitude of the point with the highest amplitude, that is, the peak, in the ISAR image, record the coordinates, and amplitude of this point. Remove this point from the original image using the CLEAN technique and then proceed to the next iteration. The process is repeated until the iteration termination condition is reached. The detailed steps of the peak extraction technique are as follows.

1) *Step 1: Initialization.* The number of iterations $l = 1$, the point set $P = \emptyset$, the solar panel image $\mathbf{I}_l(m,n)$, where m is the range index $m = 1, 2, \ldots, M$, n is the cross-range bins, $n = 1, 2, \ldots, N$, M and N are the total number of range bins and cross-range bins, respectively. The iteration termination condition (the points number reaches K or the residual energy is less than ε).
2) *Step 2:* During the lth iteration, the peak position of the image $\mathbf{I}_l(m,n)$ is searched as follows, and its position $[u_l, v_l]^T$ and the magnitude A_l is recorded. Update the location matrix $P = P \cup [u_l, v_l]^T$

$$[u_l, v_l]^T = \arg\max_{m,n} \mathbf{I}_l(m,n) \tag{53}$$

$$A_l = \mathbf{I}_l(u_l, v_l). \tag{54}$$

3) *Step 3:* Using the CLEAN algorithm to remove the point at the position $[u_l, v_l]^T$, make $l = l + 1$, return to step 2

$$\mathbf{I}_{l+1}(m,n) = \mathbf{I}_l(m,n) \odot \mathbf{W}_l(u_l, v_l) \tag{55}$$

where $\odot$ is the Hadamard product, $\mathbf{W}_l(u_l, v_l)$ is the window function of the lth iteration, which can be expressed as follows, and r_r and r_a are the radii, usually a few pixels

$$\mathbf{W}_l(u_l, v_l) = \begin{cases} 0, & |m - u_l| < r_r, |n - v_l| < r_a \\ 1, & \text{others.} \end{cases} \tag{56}$$

4) *Step 4:* Repeat steps 2 and 3 until the number of extracted points becomes K or the amplitude of the extracted point is less than ε.

After the peak extraction process, we can get the point-set location matrix $\mathbf{P}$ of the solar panel

$$\mathbf{P} = \begin{bmatrix} u_1 & u_2 & \cdots & u_K \\ v_1 & v_2 & \cdots & v_K \end{bmatrix}. \tag{57}$$

PCA is employed to extract the solar panel's principal components, the covariance matrix of $\mathbf{P}$ can be expressed as $\mathbf{C_P} = (1/K)\mathbf{P}\mathbf{P}^T$. Performing an eigenvalue decomposition on $\mathbf{C_P}$ yields $\mathbf{C_P} = \mathbf{U}\Sigma\mathbf{U}^T$, where $\mathbf{U}$ is the matrix containing eigenvectors $\tilde{\mathbf{V}}_1$ and $\tilde{\mathbf{V}}_2$, and Σ is the diagonal matrix containing the eigenvalues $\tilde{\lambda}_1$ and $\tilde{\lambda}_2$.

D. Line Feature Extraction

After acquiring the focused image, the solar panel edge line structure can be automatically extracted by using polygon fitting, morphological processing, and other relevant image processing techniques. For example, a parallelogram structural feature is extracted by the Radon transform (RT), acquiring the 2-D orientation information of the solar wing in [11] and [24]. Zhou *et al.* [11] have described the application of a parallelogram descriptor to extract solar panel edge line features, which will not be described in detail here.

E. Parameter Estimation by Parametric Optimization

After acquiring the solar panel's normal vector $\hat{\mathbf{I}}_3$, the principal component eigenvalues $\tilde{\lambda}_1$ and $\tilde{\lambda}_2$, the eigenvectors $\tilde{\mathbf{V}}_1$ and $\tilde{\mathbf{V}}_2$, and the edge line features $\tilde{\mathbf{I}}_1$ and $\tilde{\mathbf{I}}_2$, the optimization function for satellite attitude and size parameter estimation can be obtained by using (34), expressed as follows:

$$\hat{\Theta} = \arg\min_{\Theta} J(\Theta). \tag{58}$$

Since the five parameters $\Theta = [\theta, \varphi, \gamma, x_0, y_0]^T$ need to be searched simultaneously, the general exhaustive search methods, such as PSO, are slow and inaccurate. In this section, the same method as in Section IV-B, that is, the quasi-Newton iteration method based on BFGS, is adopted to perform a fast and accurate search for the five parameters. First, we need to calculate the partial derivatives of each parameter in the parameter vector Θ. The detailed procedure for gradient derivation

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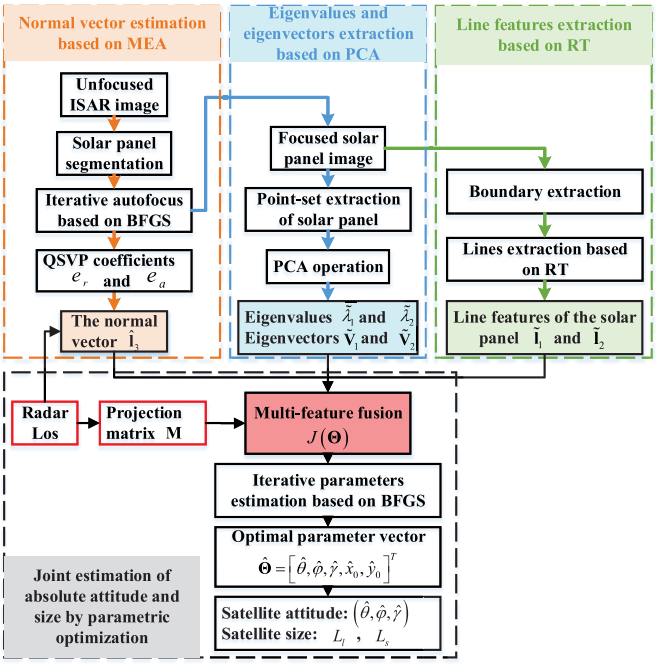


Fig. 4. Flowchart of the proposed method.

is in Appendix B. With the partial derivative expressions (72)–(91), the gradient of loss function $J(\Theta)$ for Θ is

$$\nabla \mathbf{J} = \left[\frac{\partial J(\Theta)}{\partial \theta} \quad \frac{\partial J(\Theta)}{\partial \varphi} \quad \frac{\partial J(\Theta)}{\partial \gamma} \quad \frac{\partial J(\Theta)}{\partial x_0} \quad \frac{\partial J(\Theta)}{\partial y_0} \right]^T. \tag{59}$$

With the gradient $\nabla \mathbf{J}$, the iterative process is referred to as (47)–(50) in Section IV-B, and the optimal solution of parameter $\hat{\Theta} = [\hat{\theta}, \hat{\varphi}, \hat{\gamma}, \hat{x}_0, \hat{y}_0]^T$ is obtained by fast iterative solving. With the estimated parameters $\hat{\Theta}$, the satellite attitude and size can be obtained by a simple operation, namely

$$\begin{cases} \mathbf{l}_1 = \mathbf{R}(\hat{\theta}, \hat{\varphi}, \hat{\gamma})\mathbf{n}_l \\ \mathbf{l}_2 = \mathbf{R}(\hat{\theta}, \hat{\varphi}, \hat{\gamma})\mathbf{n}_s \\ \mathbf{l}_3 = \mathbf{R}(\hat{\theta}, \hat{\varphi}, \hat{\gamma})\mathbf{n}_n, \end{cases} \quad \begin{cases} L_l = 2\hat{x}_0 \\ L_s = 2\hat{y}_0. \end{cases} \tag{60}$$

Fig. 4 gives the flowchart of the proposed method. From the figure, it can be seen that the proposed algorithm is divided into four main parts. First, the solar panel normal vector is obtained by estimating the QSVP coefficients according to ME algorithm (MEA) while acquiring the focused image. Then, based on the focused solar panel image, the solar panel point set is extracted, and the eigenvalues and eigenvectors of the point set are extracted using PCA. Meanwhile, the solar panel's long and short edge line features are extracted using the RT. Finally, the parametric optimization of the attitude and size is established by multi-feature fusion. The optimal solution of the parameters is searched rapidly based on the BFGS iteration method.

V. Experimental Analysis

Due to the lack of measured satellite observation data, this section will verify the performance of the proposed algorithm using satellite EM simulation data, which are generated by using the physical optics method [38]. Here, two experiments are designed.

1) First, the EM simulation data of the Aqua satellite under single-station observation conditions will be used to verify the effectiveness of the proposed algorithm. Here, we use multiple intervals of the observed data, each of which forms an ISAR image (conventional RD algorithm) and then each image verifies the proposed algorithm. In addition, we also performed the proposed algorithm under different signal-to-noise ratio (SNR) conditions. Eventually, the sequence images are also used to improve the accuracy and robustness of the algorithm.
2) We did extended experiments, that is, the performance of the proposed algorithm under two-station observation conditions, and compared the experimental results with the single-station condition.

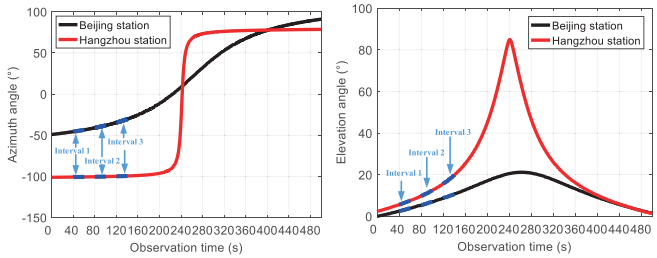
The EM simulation data are generated using radar in the Ku-band, and the main parameters of the radar are shown in Table I. The 3-D model of the Aqua satellite is shown in Fig. 5, and the model is available at nasa3d.arc.nasa.gov/models.

TABLE I
Radar Location and Parameters

Radar Location	Beijing(116.4E,39.9N,0m) Hangzhou(120.2 E, 30.3 N,0m)	
Radar Parameters	Center Frequency	16.67GHz
	PRF	50Hz
	Sample Frequency	2.4GHz
	Bandwidth	2GHz
Image Resolution	$0.075m \times 0.075m$	



Fig. 5. Three-dimensional model of the Aqua satellite.



(a) (b)

Fig. 6. (a) Azimuth angle $\alpha(t_m)$ of LOS. (b) Elevation angle $\beta(t_m)$ of LOS.

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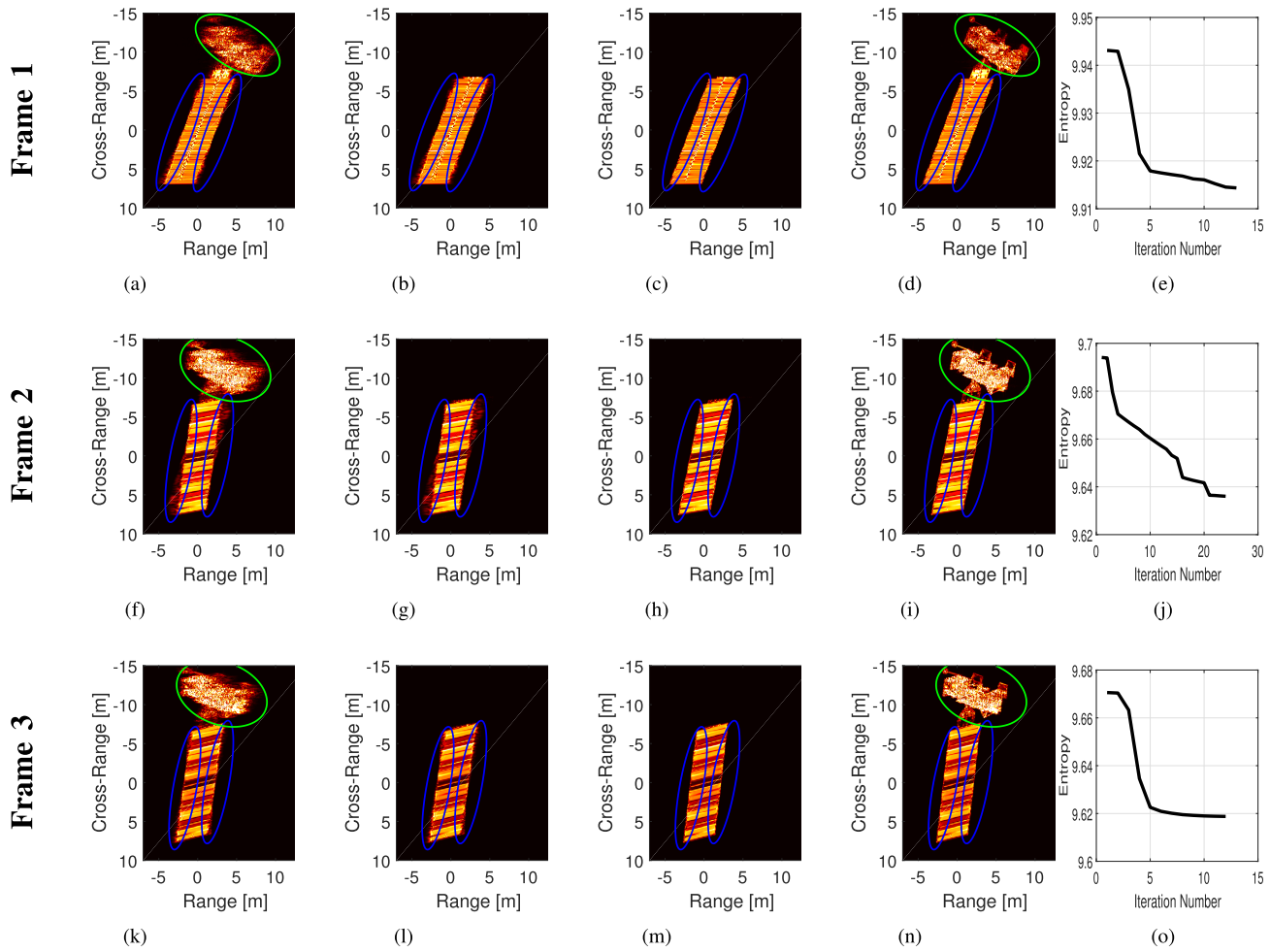


Fig. 7. MEA-based solar panel QSVPE compensation and integral image QSVPE compensation. (a), (f), and (k) Unfocused satellite integral images. (b), (g), and (l) Segmented solar panel images. (c), (h), and (m) Focused solar panel images after MEA. (d), (i), and (n) Integral focused images after compensation with estimated QSVP coefficients e_r and e_a. (e), (j), and (o) Variation curve of image entropy with the number of iterations.

The azimuth angle $\alpha(t_m)$ and elevation angle $\beta(t_m)$ of the radar LOS is generated by Satellite Tool Kits (STKs), which is consistent with the approach taken in [11] and [28].

A. Experimental Validation of Single-Station Observation

In this section, we will verify the effectiveness of the proposed method using a conventional single-station observation.

First, we generated a one-lap satellite radar observation view by STK simulation, and the observation view is shown in Fig. 6. The PO method is used to simulate the target radar echo of this lap, and the additive white Gaussian noise is added to the simulated echo, which makes an SNR of 20 dB. Three data segments are randomly selected for conventional RD imaging to obtain three unfocused ISAR images, as shown in Fig. 7(a), (f), and (k). Subsequently, the solar panels in the three images are segmented, and the solar panel segmentation results are shown in Fig. 7(b), (g), and (l). According to the proposed normal vector estimation method based on MEA, the estimated QSVP coefficients e_r and e_a are listed in Table II. The solar panels after focusing are shown in Fig. 7(c), (h), and (m), the variation curve of entropy with the number of iterations during MEA are shown in Fig. 7(e), (j), and (o).

First, it can be seen in Fig. 7(c), (h), and (m) that the solar panel images obtained after the MEA phase compensation process have significantly improved and better focus compared to Fig. 7(b), (g), and (i). As can be seen from Fig. 7(b), (g), and (i), the edges of the solar panel images have seriously defocusing (the area marked by the blue circles in the figures) due to the presence of the QSVPE. After the phase error compensation, the solar panel edge as well as the whole ISAR images are well focused and the edges contours are clearly visible. Next, the estimated QSVP coefficients are applied to compensate the QSVPE for the whole image, and it can be seen that the entire image also has a significantly better focusing performance. The estimated QSVP coefficients e_r and e_a are brought into (21) to estimate the normal vectors of the solar panel. As seen in Table II, the estimated normal vectors of the three images are basically the same. The error between them and the true normal vectors is only a few degrees, which verifies the accuracy of the estimated normal vectors.

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TABLE II
ESTIMATED QSVPE COEFFICIENTS AND THE NORMAL VECTORS

	Frame 1	Frame 2	Frame 3
Entropy before MEA	9.943	9.694	9.671
Entropy after MEA	9.914	9.636	9.619
$e_r(\times 10^{-5})$	-8.526	-6.752	-5.630
$e_a(\times 10^{-3})$	-3.411	-5.339	-6.170
Estimated normal vector	(0.54,0.46,0.71)	(0.57,0.40,0.72)	(0.58,0.46,0.67)
Real normal vector	(0.54,0.45,0.72)		
Error [°]	0.15°	3.84°	2.88°

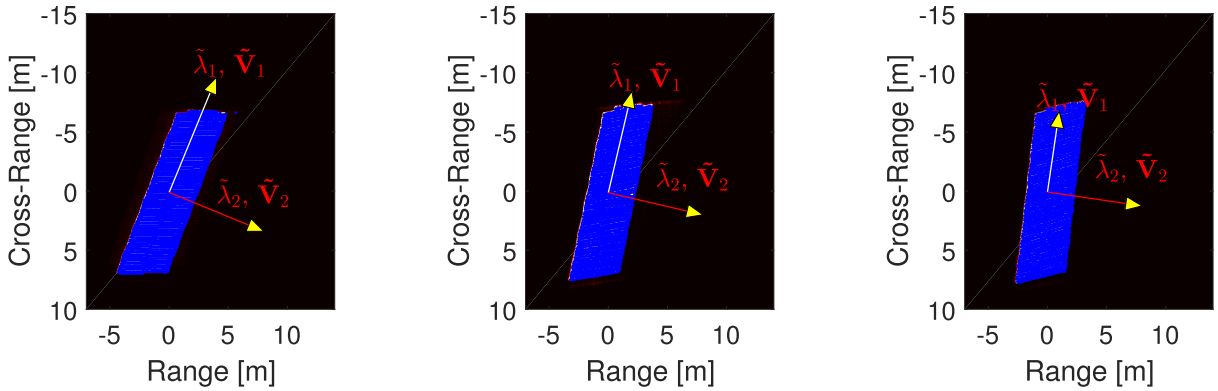


Fig. 8. Solar panel's eigenvalues and eigenvectors extraction results. (a) Frame 1. (b) Frame 2. (c) Frame 3.

TABLE III
EXTRACTED EIGENVALUES AND EIGENVECTORS

	Frame 1	Frame 2	Frame 3
$\tilde{\lambda}_1$	17.68	18.29	18.58
$\tilde{\lambda}_2$	1.24	1.33	1.34
$\tilde{\mathbf{V}}_1$	(0.92,0.38)	(0.97,0.23)	(0.99,0.15)
$\tilde{\mathbf{V}}_2$	(-0.38,0.92)	(-0.23,0.97)	(-0.15,0.99)

Then, after acquiring the focused solar panel image, the scattering point extraction method based on the peak extraction technique proposed in Section IV-C is used to extract the solar panel's point set, and the extracted point set is shown in Fig. 8. Based on the extracted point sets, the eigenvalues and the eigenvectors of the solar panel point sets are extracted using PCA, as shown in Fig. 8(a)–(c). The eigenvalues and eigenvectors are listed in Table III.

The edge line structure is extracted using morphological filter-based boundary extraction and RT transform based on the acquired focused solar panel image. The extracted long and short edges of the solar panel are shown in Fig. 9.

After acquiring the solar panel normal vectors $\hat{\mathbf{I}}_3$, eigenvalues $\tilde{\lambda}_1$ and $\tilde{\lambda}_2$, the eigenvectors $\tilde{\mathbf{V}}_1$ and $\tilde{\mathbf{V}}_2$, the line features $\tilde{\mathbf{I}}_1$ and $\tilde{\mathbf{I}}_2$, and the projection lengths L_1 and L_2, the BFGS-based gradient iteration method is used to solve for the satellite attitude and size parameters based on the established absolute attitude and size estimation optimization function. The variation curve of the fitness function value $J(\Theta)$ with the number of iterations during the iterative process is shown in Fig. 10, and we similarly plot the variation of the attitude and size errors with the number of iterations. The absolute attitude and size estimated from the three selected observation intervals are listed in Table IV. It should be emphasized that "Frame 1," "Frame 2," and "Frame 3" refer to the results of the attitude and size estimation using a single ISAR image, respectively. "Joint estimation" represents the results of parameter estimation using three images jointly. Furthermore, we can combine three images for parameter estimation. Considering that (34) is established for a single image, but the cost function of multiple images conforms to the linear summation characteristic, that is, the cost function for joint estimation attitude and size for multiple images is

$$J_M(\Theta) = \sum_{m=1}^{M} J^m(\Theta) \tag{61}$$

where m is the frame index, $m = 1, 2, \ldots, M$, and M is the total image that used to estimate the attitude and size

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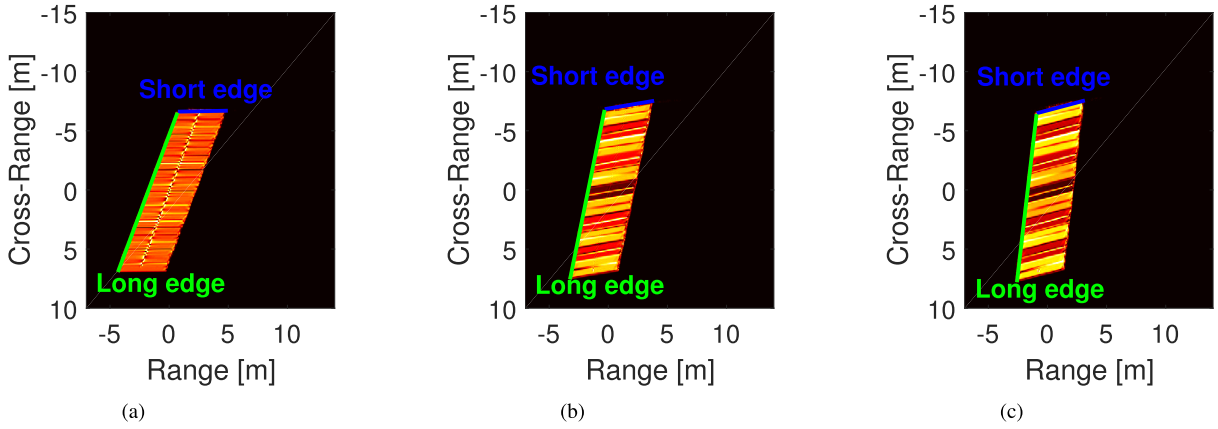


Fig. 9. Solar panel's long and short edges extraction results. (a) Frame 1. (b) Frame 2. (c) Frame 3.

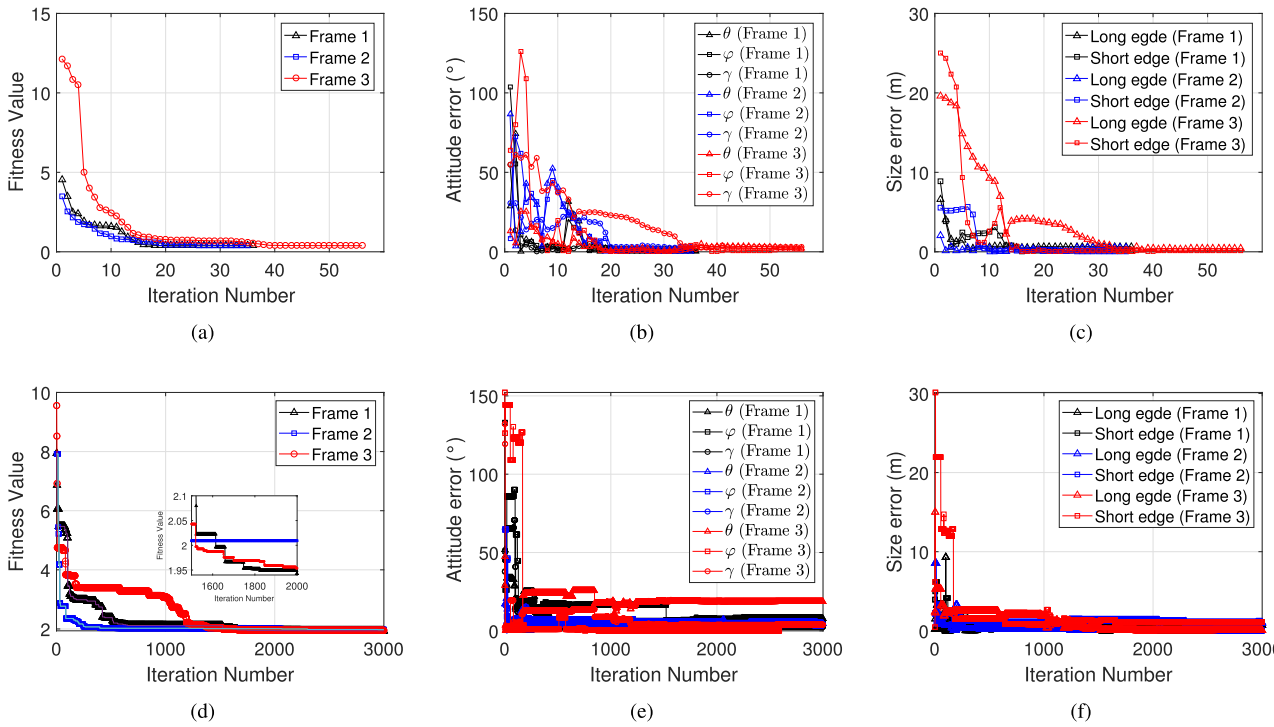


Fig. 10. Iteration curves during attitude and size estimation via using the proposed method and PSO. (a) Proposed method. (d) PSO algorithm: variation of the optimization function values with the number of iterations. (b) Proposed method. (e) PSO algorithm: variation curve of attitude errors with the number of iterations. (c) Proposed method. (f) PSO algorithm: variation curve of size errors with the number of iterations.

parameters. The gradient of $J_M(\Theta)$ also conforms to the linear summation characteristic. Thus, the attitude and size can be estimated by associating multiple images to improve the estimation accuracy.

The curve of the iterative process shows that the proposed gradient-based optimization method can search for the optimal values of the five parameters rapidly and accurately. The estimated attitude and size results are listed in Table IV. From the table, we can see that the attitude error is within an acceptable 5° and the size error is within 1 m. After estimating the attitude and size jointly by associating the three images, it can be seen that the attitude and size errors estimated using three images jointly are close to the average of the results estimated using only a single image, which makes the estimation results more robust. Compared with the method proposed in the literature [28], the proposed method is more effective because the method proposed in [28] can only estimate the satellite solar panel normal vector. In contrast, the proposed method can estimate both attitude and size.

In the following, we give the computational time comparison between the PSO algorithm and the proposed method. Here, we take the Beijing station observation with the target in pose I as an example. The iteration curves using the two search algorithms are shown in Fig. 12(d)–(f). Fig. 12(a) and (d) are the curves of the variation of the fitness function values with the number of iterations during the search of the proposed algorithm and the PSO algorithm. From Fig. 12(a) and (d), it can be seen that the number of iterations required for convergence of the two algorithms differs greatly. The proposed

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TABLE IV
ATTITUDE AND SIZE ESTIMATION RESULTS OF AQUA SATELLITE AT POSE I

Pose I			True value	Frame 1		Frame 2		Frame 3		Joint estimation	
				Estimated	Error	Estimated	Error	Estimated	Error	Estimated	Error
Beijing Station	Attitude [°]	θ	40°	39.11°	0.89°	37.8°	2.18°	38.76°	1.24°	38.71°	1.29°
		φ	45°	44.53°	0.47°	45.19°	0.19°	45.72°	0.72°	44.88°	0.12°
		γ	0°	0.39°	0.39°	1.81°	1.81°	1.17°	1.17°	1.10°	1.10°
	Size [m]	Long edge	15m	15.41m	0.41m	15.30m	0.30m	15.24m	0.24m	15.36m	0.36m
		Short edge	6.16m	6.06m	0.10m	6.66m	0.50m	6.49m	0.33m	6.46m	0.30m
Hangzhou Station	Attitude [°]	θ	40°	41.47°	1.47°	44.76°	4.76°	35.83°	4.17°	41.13°	1.13°
		φ	45°	46.50°	1.50°	43.95°	1.05°	46.18°	1.18°	45.27°	0.27°
		γ	0°	0.61°	0.61°	0.66°	0.66°	1.56°	1.56°	0.14°	0.14°
	Size [m]	Long edge	15m	15.09m	0.09m	15.73m	0.73m	15.24m	0.24m	15.23m	0.23m
		Short edge	6.16m	6.88m	0.72m	6.16m	0.00m	6.98m	0.82m	6.57m	0.41m
Two-Station	Attitude [°]	θ	40°	39.18°	0.81°	38.34°	1.66°	37.13°	2.87°	39.22°	0.78°
		φ	45°	45.15°	0.15°	45.60°	0.60°	42.21°	2.79°	45.63°	0.63°
		γ	0°	0.62°	0.62°	2.48°	2.48°	2.02°	2.02°	0.33°	0.33°
	Size[m]	Long edge	15m	15.25m	0.25m	15.04m	0.04m	15.23m	0.23m	15.19m	0.19m
		Short edge	6.16m	5.94m	0.22m	6.63m	0.47m	6.80m	0.64m	6.06m	0.10m

method reaches convergence in about 20 iterations, while the PSO-based algorithm requires thousands of iterations for convergence.

In addition, we give the time required for a single iteration of both algorithms, and the computational time required for convergence. Table V shows the computation time of these two methods obtained by running MATLAB codes on a personal computer with an Intel Core i5 3.30-GHz processer and 8-GB memory. As one can note, compared with PSO, the proposed method needs fewer iterations and computation time to achieve convergence. The reason is that, compared with the gradient-based search method, PSO is an exhaustive search method. Unlike the gradient-based search method, PSO does not search in the optimal direction during each iteration, resulting in slow convergence and imprecise search. However, the proposed method requires fewer iterations to reach the optimal solution, saving a lot of computation time.

TABLE V
COMPUTATION TIME OF ONE ITERATION AND CONVERGENCE FOR PSO AND THE PROPOSED METHOD

	Proposed method	PSO
One iteration [s]	2.4×10^{-3}	6.26×10^{-4}
Iteration number for convergence	20	1500
Computation time for convergence [s]	0.048	0.939

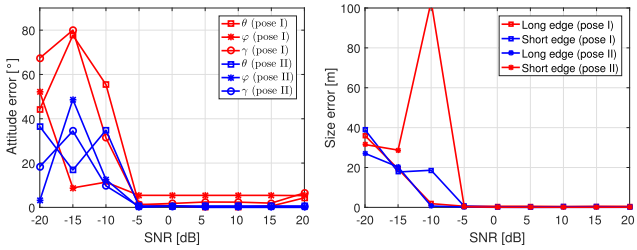
To better reflect the effectiveness of the proposed algorithm, we also did experiments when the satellite was in another

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TABLE VI
ATTITUDE AND SIZE ESTIMATION RESULTS OF AQUA SATELLITE AT POSE II

Pose II			True value	Frame 1		Frame 2		Frame 3		Joint estimation	
				Estimated	Error	Estimated	Error	Estimated	Error	Estimated	Error
Beijing Station	Attitude [°]	θ	90°	89.76°	0.24°	90.02°	0.02°	92.21°	2.21°	89.23°	0.77°
		φ	65°	67.20°	2.2°	67.91°	2.91°	68.20°	3.2°	67.90°	2.9°
		γ	0°	0.83°	0.83°	1.57°	1.57°	1.82°	1.82°	0.89°	0.89°
	Size [m]	Long edge	15m	15.65m	0.65m	15.53m	0.53m	15.38m	0.38m	15.40m	0.40m
		Short edge	6.16m	6.34m	0.18m	6.23m	0.07m	6.29m	0.13m	6.28m	0.12m
Hangzhou Station	Attitude [°]	θ	90°	90.15°	0.15°	90.43°	0.43°	90.06°	0.06°	89.89°	0.11°
		φ	65°	66.23°	1.23°	64.51°	0.49°	63.08°	1.92°	64.50°	0.50°
		γ	0°	0.24°	0.24°	0.91°	0.91°	2.43°	2.43°	1.39°	1.39°
	Size [m]	Long edge	15m	15.22m	0.22m	15.44m	0.44m	15.78m	0.78m	15.48m	0.48m
		Short edge	6.16m	6.38m	0.22m	6.37m	0.21m	6.30m	0.14m	6.30m	0.14m
Two-Station	Attitude [°]	θ	90°	89.40°	0.60°	90.06°	0.06°	89.83°	0.17°	89.97°	0.03°
		φ	65°	64.36°	0.64°	64.99°	0.01°	66.90°	1.90°	66.38°	1.38°
		γ	0°	0.51°	0.51°	1.57°	1.57°	2.34°	2.34°	0.44°	0.44°
	Size[m]	Long edge	15m	15.53m	0.53m	15.36m	0.36m	15.14m	0.14m	15.17m	0.17m
		Short edge	6.16m	6.28m	0.12m	6.37m	0.21m	6.30m	0.14m	6.28m	0.12m

pose, and the experimental results are shown in Table VI. The results of the attitude and size estimation in two poses show that the proposed method is effective. Previous experiments validate the feasibility and accuracy of the proposed algorithm. Furthermore, to investigate the proposed algorithm's sensitivity against random noise, the attitude and size estimation experiment is repeated under different additive white Gaussian noise conditions. In the experiment, the SNR is from -20 to 20 dB. The curves of attitude and size estimation results with SNR are shown in Fig. 11. The figure shows that the proposed method has good accuracy despite the low SNR (like not lower than -5 dB), which reflects the noise robustness of the proposed algorithm.



(a) (b)

Fig. 11. Attitude and size estimation error with SNR. (a) Curve of the attitude estimation error of the satellite with respect to the SNR. (b) Curve of the size estimation error of the satellite with respect to the SNR.

B. Experimental Validation of Multi-Station Observation

The experimental results for the case of single-station observation are given in Section V-A. However, in some cases, the single-station observation has some limitations. In some views, the projection of the satellite solar panel on the ISAR image will be a straight line or a very

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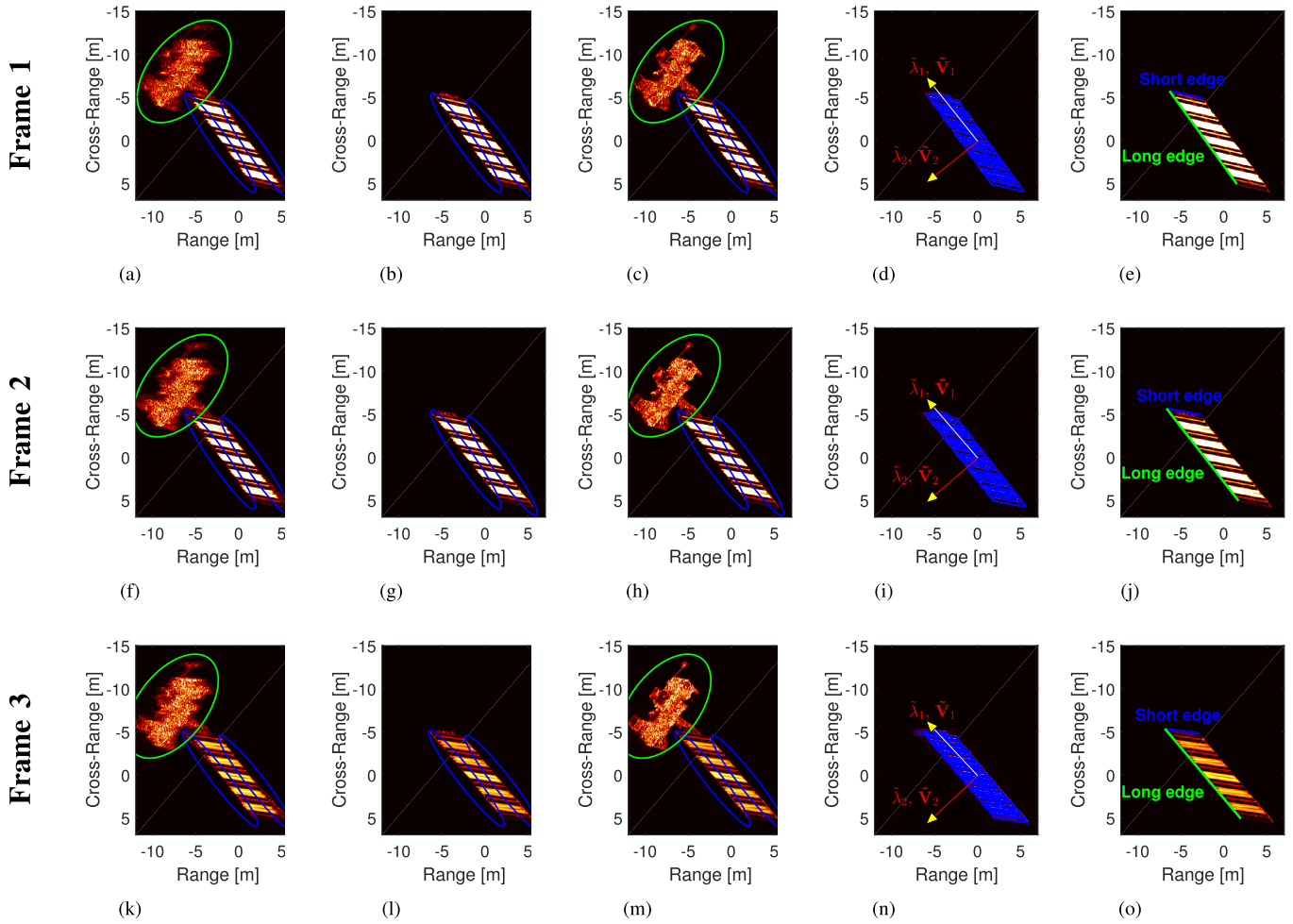


Fig. 12. Multi-feature extraction results of satellite images observed at Hangzhou station. (a), (f), and (k) Unfocused satellite integral images. (b), (g), and (l) Focused solar panel images after MEA. (c), (h), and (m) Integral focused images after compensation with estimated QSVP coefficients e_r and e_a. (d), (i), and (n) Solar panel's eigenvalues and eigenvectors extraction results. (e), (j), and (o) Solar panel's long and short edges extraction results.

narrow rectangle. The estimation error of the proposed algorithm will be significantly increased. To avoid this situation, employing multi-station observations is an effective way to make the satellite imaging projection plane different at the exact moment, increasing the available projection information and making the proposed method more robust.

This section will add a new radar station with the same parameters as in Table I in Hangzhou (120.2E, 30.3N, 0 m). In this case, two radars observe the satellite simultaneously, and three time periods are also chosen. Each radar acquires three images, and the satellite attitude and size are estimated jointly using (62). The multi-feature extraction results are shown in Fig. 12. The estimation results are shown in Table VI.

As can be seen from Table VI, more accurate results can be obtained by using two-station observations for attitude and size estimation compared to single-station observations. The results of the two-station observations are more evenly in error compared to the single-station ones, and in some cases are even more accurate than in the single-station ones.

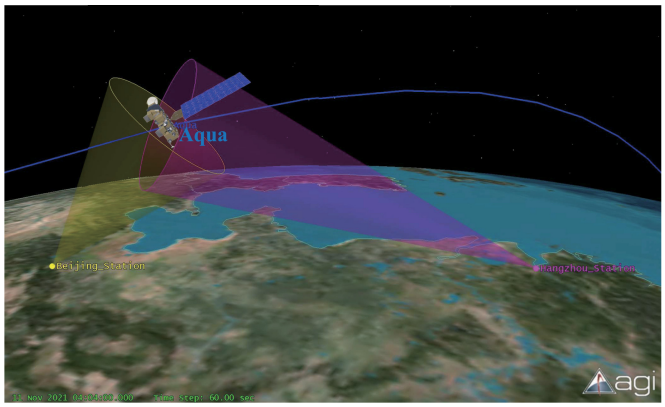


Fig. 13. Two-station observation simulated by STK.

This is due to the fact that compared to single-station cases, multi-station observations can acquire information from multiple views of the satellite at the same moment, providing richer structural features and making the estimates more robust.

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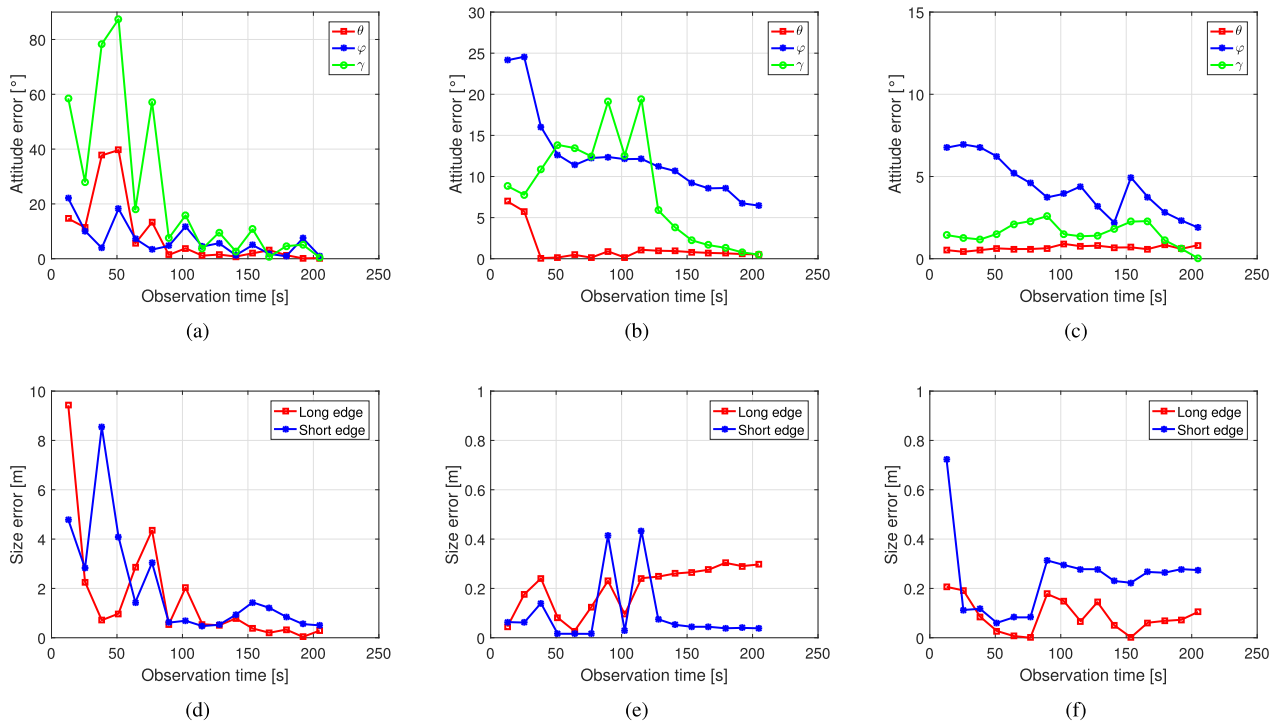


Fig. 14. Comparison of the proposed method with the long-time observation-based algorithm at pose I (observed by the Beijing station). (a) and (d) Long-time observation algorithm based on feature extraction of the solar panel via PCA (algorithm proposed in [25]). (b) and (e) Long-time observation algorithm based on solar panel edge line feature (algorithm proposed in [11]). (c) and (f) Algorithm proposed in this article. (a)–(c) Variation curves of the attitude error with observation time for different algorithms. (d)–(f) Variation curves of size estimation error with observation time for a different algorithm.

C. Comparison With Long-Time Observation-Based Methods

To reflect the advantages of the proposed algorithm, in this section, the proposed algorithm is compared with the existing methods based on long-time observation data. A satellite attitude and size estimation method based on the line structure feature of the solar panel edge, that is, the line feature among the multiple features of the proposed method, is presented in literature [11]. The method requires long-time observation to acquire satellite sequence images and uses the sequence image projection difference to estimate the satellite attitude and size based on the imaging projection principle. Literature [25] proposes a method based on solar panel eigenvalues and eigenvectors (i.e., solar panel features in the proposed multi-features in this article), which also requires multiple images acquired by long-time observation. Take the satellite observation at pose I, the Beijing station as an example. The curves in Fig. 14 compare the estimation errors of these two methods with the proposed method. It can be seen that the estimation errors of the methods presented in the literature [11] and [25] are larger when there are few observation data, and the attitude errors reach tens of degrees. In contrast, the proposed method is still effective with few observation data, and the attitude estimation errors are within 10°. With the increase of observation time, the estimation errors of the methods proposed in the literature [11] and [25] gradually decrease. The comparison shows that the proposed method has a high attitude and size estimation accuracy in both the case of sufficient data and few data. In contrast, the two compared methods will fail in the case of few observation data due to their reliance on a large amount of data from a long period of observations.

D. Some Discussions

The experimental analysis shows that the proposed method is effective in practical applications, the proposed algorithm does not depend on long-time radar observations, and the robustness of the proposed algorithm can be improved under multi-station conditions. However, during the experimental process, we also find that the proposed algorithm has some limitations. First, the proposed method relies on the extraction accuracy of multiple features. Since the QSVPE coefficients estimation is obtained based on the refocusing of the unfocused ISAR images, its estimation accuracy depends on the focusing quality of ISAR images. In addition, the segmentation of solar panels also has an impact on the QSVPE coefficients estimation. In the subsequent study, we will analyze the mechanism of these effects and how to estimate the QSVPE coefficients robustly in more depth. Second, the solar panel eigenvalue, eigenvector extraction, and edge line feature extraction are affected by the integrity of the solar panel, which requires high imaging quality. Despite some shortcomings, the proposed algorithm has obvious advantages, that is, only a single image is required for joint estimation of satellite attitude and size, and only a small amount of observation data is required for practical applications, which provides a feasible idea for subsequent analysis of satellite state with limited data.

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VI. Conclusion

In this article, a joint satellite attitude and size estimation method based on the multi-feature fusion of a single ISAR image is proposed to estimate the absolute attitude and size of a satellite under the condition of limited observational data. To a certain extent, the method requires only one ISAR image of the satellite to achieve attitude and size estimation. The method utilizes rectangular solar panels commonly equipped by satellites. It combines the projection principle of ISAR imaging to establish the mapping relationship between multi-features such as QSVPE features, line features, and principal component features in the solar panel ISAR images and satellite attitude and size parameters. The mapping relationship between the multi-features and the attitude and size parameters is combined to establish a parametric estimation optimization function. The BFGS-based quasi-Newton method is used for a fast search of attitude and size parameters to achieve high accuracy estimation of the absolute attitude and size of the satellite. The effectiveness of the proposed algorithm is verified by experiments with different attitudes under single/multi-station observation conditions. The comparison with existing methods based on long-time observation data proves that the proposed method does not depend on a large amount of observation data and demonstrates the effectiveness of the proposed algorithm.

Appendix A

The mapping relationship between the O-XYZ coordinate system and the ISAR imaging plane can be expressed as (17). Considering that $\mathbf{M} = \begin{bmatrix} \mathbf{i}^T(t_0)/\Delta r \\ \mathbf{i}'^T(t_0)/\Delta f_a \end{bmatrix}$ and $\hat{\mathbf{M}} = \mathbf{M}\mathbf{R}(\theta, \varphi, \gamma)$, assume that $\hat{\mathbf{M}} = \begin{bmatrix} *20c\hat{\mathbf{i}} \\ \hat{\mathbf{j}} \end{bmatrix}$, where

$$\begin{cases} \hat{\mathbf{i}}^T = \left(\mathbf{i}^T(t_0)/\Delta r\right)\mathbf{R}(\theta, \varphi, \gamma) \\ \hat{\mathbf{j}}^T = \left(\mathbf{i}'^T(t_0)/\Delta f_a\right)\mathbf{R}(\theta, \varphi, \gamma). \end{cases} \tag{62}$$

Assume that $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ can be expressed as

$$\begin{cases} \hat{\mathbf{i}} = [i_1, i_2, i_3]^T \\ \hat{\mathbf{j}} = [j_1, j_2, j_3]^T. \end{cases} \tag{63}$$

In the O-XYZ coordinate system, the coordinates of the point set $\mathbf{Q}$ can be expressed as

$$\mathbf{Q} = \begin{bmatrix} x_1 & x_2 & \cdots & x_K \\ y_1 & y_2 & \cdots & y_K \\ z_1 & z_2 & \cdots & z_K \end{bmatrix}. \tag{64}$$

Then, according to (17), the point set of solar panel on the ISAR image is

$$\hat{\mathbf{P}} = \hat{\mathbf{M}}\mathbf{Q} = \begin{bmatrix} \hat{\mathbf{i}} \\ \hat{\mathbf{j}} \end{bmatrix} \begin{bmatrix} x_1 & x_2 & \cdots & x_K \\ y_1 & y_2 & \cdots & y_K \\ z_1 & z_2 & \cdots & z_K \end{bmatrix} = \begin{bmatrix} u_1 & u_2 & \cdots & u_K \\ v_1 & v_2 & \cdots & v_K \end{bmatrix}. \tag{65}$$

According to (65), we have

$$\begin{cases} u_i = i_1 x_i + i_2 y_i + i_3 z_i \\ v_i = j_1 x_i + j_2 y_i + j_3 z_i. \end{cases} \tag{66}$$

Considering $z_i = 0$ in the O-XYZ coordinate system, so the variances of u and v, as well as their covariance, can be expressed as

$$\begin{cases} D_u = i_1^2 D_x + i_2^2 D_y \\ D_v = j_1^2 D_x + j_2^2 D_y \\ D_{uv} = i_1 j_1 D_x + i_2 j_2 D_y \end{cases} \tag{67}$$

where D_x and D_y contain the solar panel size x_0 and y_0, as shown in (25). Bringing (67) into (28) yields

$$\mathbf{C}_{\hat{\mathbf{P}}} = \begin{bmatrix} D_u & D_{uv} \\ D_{vu} & D_v \end{bmatrix}. \tag{68}$$

The SVD decomposition of (68) yields the eigenvalues and the eigenvectors as

$$\begin{cases} \lambda_1(\Theta) = \dfrac{D_u + D_v + \sqrt{(D_u + D_v)^2 - 4\left(D_u D_v - D_{uv}^2\right)}}{2} \\ \lambda_2(\Theta) = \dfrac{D_u + D_v - \sqrt{(D_u + D_v)^2 - 4\left(D_u D_v - D_{uv}^2\right)}}{2} \end{cases} \tag{69}$$

$$\mathbf{V}_1(\Theta) = \begin{bmatrix} \dfrac{1}{\sqrt{1 + \left(\frac{\lambda_1 - D_u}{D_{uv}}\right)^2}} \\ \dfrac{1}{\sqrt{1 + \left(\frac{D_{uv}}{\lambda_1 - D_u}\right)^2}} \end{bmatrix}, \quad \mathbf{V}_2(\Theta) = \begin{bmatrix} \dfrac{1}{\sqrt{1 + \left(\frac{\lambda_2 - D_u}{D_{uv}}\right)^2}} \\ \dfrac{1}{\sqrt{1 + \left(\frac{D_{uv}}{\lambda_2 - D_u}\right)^2}} \end{bmatrix}. \tag{70}$$

Appendix B

Considering the optimization function of (34), which consists of $J_1(\theta, \varphi, \gamma)$, $J_2(\Theta)$ and $J_3(\Theta)$, solving for the partial derivatives of each parameter for $J(\Theta)$, one can find the partial derivatives of $J_1(\theta, \varphi, \gamma)$, $J_2(\Theta)$, and $J_3(\Theta)$ with respect to the five parameters, respectively. First, starting from $J_1(\theta, \varphi, \gamma)$, as shown in (22), since it contains three parameters, that is, θ, φ, and γ, the derivatives of $J_1(\theta, \varphi, \gamma)$ with respect to these three parameters are

$$\begin{aligned} \frac{\partial J_1(\theta, \varphi, \gamma)}{\partial \Phi} &= -\frac{1}{\sqrt{1 - \left(\mathbf{l}_3^T(\theta, \varphi, \gamma) \cdot \hat{\mathbf{l}}_3\right)^2}} \cdot \frac{\partial\left(\mathbf{l}_3^T(\theta, \varphi, \gamma) \cdot \hat{\mathbf{l}}_3\right)}{\partial \Phi} \\ &= -\frac{1}{\sqrt{1 - \left(\mathbf{l}_3^T(\theta, \varphi, \gamma) \cdot \hat{\mathbf{l}}_3\right)^2}} \frac{\partial \mathbf{R}(\theta, \varphi, \gamma)}{\partial \Phi} \mathbf{n}_n \cdot \hat{\mathbf{l}}_3 \end{aligned} \tag{71}$$

where $\mathbf{l}_3 = \mathbf{R}(\theta, \varphi, \gamma)\mathbf{n}_n$, $\mathbf{n}_n = [0, 0, 1]^T$, which were introduced in Section III. Φ denotes the angle variables $(\theta, \varphi, \gamma)$. $(\partial \mathbf{R}(\theta, \varphi, \gamma)/\partial \Phi)$ is given as follows:

$$\begin{cases} \dfrac{\partial \mathbf{R}(\theta, \varphi, \gamma)}{\partial \theta} = \dfrac{\partial \mathbf{R}_z(\theta)}{\partial \theta} \mathbf{R}_y(\varphi) \mathbf{R}_x(\gamma) \\ \dfrac{\partial \mathbf{R}(\theta, \varphi, \gamma)}{\partial \varphi} = \mathbf{R}_z(\theta) \dfrac{\partial \mathbf{R}_y(\varphi)}{\partial \varphi} \mathbf{R}_x(\gamma) \\ \dfrac{\partial \mathbf{R}(\theta, \varphi, \gamma)}{\partial \gamma} = \mathbf{R}_z(\theta) \mathbf{R}_y(\varphi) \dfrac{\partial \mathbf{R}_x(\gamma)}{\partial \gamma} \end{cases} \tag{72}$$

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where

$$\frac{\partial \mathbf{R}_z(\theta)}{\partial \theta} = \begin{bmatrix} -\sin\theta & -\cos\theta & 0 \\ \cos\theta & -\sin\theta & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{73}$$

$$\frac{\partial \mathbf{R}_y(\varphi)}{\partial \varphi} = \begin{bmatrix} -\sin\varphi & 0 & \cos\varphi \\ 0 & 0 & 0 \\ -\cos\varphi & 0 & -\sin\varphi \end{bmatrix} \tag{74}$$

$$\frac{\partial \mathbf{R}_x(\gamma)}{\partial \gamma} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\sin\gamma & -\cos\gamma \\ 0 & \cos\gamma & -\sin\gamma \end{bmatrix}. \tag{75}$$

Now, calculate the derivatives of $J_2(\Theta)$ with respect to each parameter. Considering that (30) contains five parameters, the derivatives are solved for each of these five parameters. Considering the eigenvalues in (69) and the eigenvectors in (70) are both composed of D_u, D_v, and D_uv, the partial derivative of an arbitrary variable can be expressed as

$$\frac{\partial J_2(\Theta)}{\partial \Theta} = \frac{\partial J_2(\Theta)}{\partial D_u}\frac{\partial D_u}{\partial \Theta} + \frac{\partial J_2(\Theta)}{\partial D_v}\frac{\partial D_v}{\partial \Theta} + \frac{\partial J_2(\Theta)}{\partial D_{uv}}\frac{\partial D_{uv}}{\partial \Theta} \tag{77}$$

where Θ represents one of the five variables $(\theta, \varphi, \gamma, x_0, y_0)$. The partial derivatives of D_u, D_v, and D_uv can be expressed as

$$\frac{\partial J_2(\Theta)}{\partial \varsigma} = \sum_{i=1}^{2} \frac{\partial\left|\lambda_i(\Theta) - \tilde{\lambda}_i\right|}{\partial \varsigma} - \sum_{j=1}^{2} \frac{\frac{\partial\left(\mathbf{V}_j^T(\Theta)\cdot\tilde{\mathbf{V}}_j\right)}{\partial \varsigma}}{\sqrt{1-\left(\mathbf{V}_j^T(\Theta)\cdot\tilde{\mathbf{V}}_j\right)^2}} \tag{78}$$

where ς represents D_u, D_v, and D_uv. $((\partial|\lambda_i(\Theta) - \tilde{\lambda}_i|)/\partial\varsigma)$ can be expressed as

$$\frac{\partial\left|\lambda_i(\Theta) - \tilde{\lambda}_i\right|}{\partial \varsigma} = \frac{\lambda_i(\Theta) - \tilde{\lambda}_i}{\left|\lambda_i(\Theta) - \tilde{\lambda}_i\right|} \cdot \frac{\partial \lambda_i(\Theta)}{\partial \varsigma} \tag{79}$$

$(\partial\lambda_1(\Theta)/\partial\varsigma)$ and $(\partial\lambda_2(\Theta)/\partial\varsigma)$ are given as follows:

$$\begin{cases} \dfrac{\partial \lambda_1(\Theta)}{\partial D_u} = \dfrac{\partial \lambda_2(\Theta)}{\partial D_v} = \dfrac{1}{2} + \dfrac{D_u - D_v}{2\sqrt{(D_u - D_v)^2 + 4D_{uv}^2}} \\ \dfrac{\partial \lambda_1(\Theta)}{\partial D_v} = \dfrac{\partial \lambda_2(\Theta)}{\partial D_u} = \dfrac{1}{2} + \dfrac{D_v - D_u}{2\sqrt{(D_u - D_v)^2 + 4D_{uv}^2}} \\ \dfrac{\partial \lambda_1(\Theta)}{\partial D_{uv}} = -\dfrac{\partial \lambda_2(\Theta)}{\partial D_{uv}} = \dfrac{2D_{uv}}{\sqrt{(D_u - D_v)^2 + 4D_{uv}^2}} \end{cases} \tag{80}$$

$((\partial(\mathbf{V}_j^T(\Theta)\cdot\tilde{\mathbf{V}}_j))/\partial\varsigma)$ can be expressed as

$$\frac{\partial\left(\mathbf{V}_j^T(\Theta)\cdot\tilde{\mathbf{V}}_j\right)}{\partial \varsigma} = \frac{\partial \mathbf{V}_j^T(\Theta)}{\partial \varsigma}\cdot\tilde{\mathbf{V}}_j. \tag{81}$$

Considering the complexity of deriving $\mathbf{V}_j^T(\Theta)$, here we define the angle between the eigenvectors and the Doppler axis in ISAR image (i.e., $\mathbf{f}_a$-axis) as ϕ_j, which can be represented as

$$\phi_j(\Theta) = \arctan\frac{\mathbf{V}_j(2)}{\mathbf{V}_j(1)} = \arctan\frac{\lambda_j - D_u}{D_{uv}}. \tag{82}$$

Then the eigenvectors can be re-expressed as

$$\mathbf{V}_1(\Theta) = \begin{bmatrix} \cos\phi_1(\Theta) \\ \sin\phi_1(\Theta) \end{bmatrix}, \quad \mathbf{V}_2(\Theta) = \begin{bmatrix} \cos\phi_2(\Theta) \\ \sin\phi_2(\Theta) \end{bmatrix}. \tag{83}$$

In this case, the derivative of $\mathbf{V}_j^T(\Theta)$ is

$$\frac{\partial \mathbf{V}_j^T(\Theta)}{\partial \varsigma} = \begin{bmatrix} -\sin\phi_j(\Theta) \\ \cos\phi_j(\Theta) \end{bmatrix} \cdot \frac{\partial \phi_j(\Theta)}{\partial \varsigma}. \tag{84}$$

$(\partial\phi_1(\Theta)/\partial\varsigma)$ and $(\partial\phi_2(\Theta)/\partial\varsigma)$ are given as follows:

$$\begin{cases} \dfrac{\partial \phi_1(\Theta)}{\partial D_u} = \dfrac{\partial \phi_2(\Theta)}{\partial D_u} = \dfrac{-D_{uv}}{4D_{uv}^2 + (D_u - D_v)^2} \\ \dfrac{\partial \phi_1(\Theta)}{\partial D_v} = \dfrac{\partial \phi_2(\Theta)}{\partial D_v} = \dfrac{D_{uv}}{4D_{uv}^2 + (D_u - D_v)^2} \\ \dfrac{\partial \phi_1(\Theta)}{\partial D_{uv}} = \dfrac{\partial \phi_2(\Theta)}{\partial D_{uv}} = \dfrac{D_u - D_v}{4D_{uv}^2 + (D_u - D_v)^2}. \end{cases} \tag{85}$$

As shown in (77), the partial derivatives of the five variables differ from each other in $(\partial D_u/\partial\Theta)$, $(\partial D_v/\partial\Theta)$, and $(\partial D_{uv}/\partial\Theta)$. Combined with (67) and (25), $(\partial D_u/\partial\Theta)$, $(\partial D_v/\partial\Theta)$, and $(\partial D_{uv}/\partial\Theta)$ can be expressed as follows:

$$\begin{cases} \dfrac{\partial D_u}{\partial \Phi} = \dfrac{2i_1x_0^2}{3}\dfrac{\partial i_1}{\partial \Phi} + \dfrac{2i_2y_0^2}{3}\dfrac{\partial i_2}{\partial \Phi} \\ \dfrac{\partial D_v}{\partial \Phi} = \dfrac{2j_1x_0^2}{3}\dfrac{\partial j_1}{\partial \Phi} + \dfrac{2j_2y_0^2}{3}\dfrac{\partial j_2}{\partial \Phi} \\ \dfrac{\partial D_{uv}}{\partial \Phi} = \dfrac{x_0^2}{3}\left(\dfrac{\partial i_1}{\partial \Phi}j_1 + i_1\dfrac{\partial j_1}{\partial \Phi}\right) + \dfrac{y_0^2}{3}\left(\dfrac{\partial i_2}{\partial \Phi}j_2 + i_2\dfrac{\partial j_2}{\partial \Phi}\right) \end{cases} \tag{86}$$

$$\begin{cases} \dfrac{\partial D_u}{\partial x_0} = \dfrac{2i_1^2x_0}{3}, \quad \dfrac{\partial D_u}{\partial y_0} = \dfrac{2i_2^2y_0}{3} \\ \dfrac{\partial D_v}{\partial x_0} = \dfrac{2j_1^2x_0}{3}, \quad \dfrac{\partial D_v}{\partial y_0} = \dfrac{2j_2^2y_0}{3} \\ \dfrac{\partial D_{uv}}{\partial x_0} = \dfrac{2i_1j_1x_0}{3}, \quad \dfrac{\partial D_{uv}}{\partial y_0} = \dfrac{2i_2j_2y_0}{3}. \end{cases} \tag{87}$$

Finally, the derivatives of $J_3(\Theta)$ with respect to the five parameters are calculated. Considering (32) and (33), $\hat{\mathbf{I}}_1(\theta, \varphi, \gamma) = \hat{\mathbf{M}}\mathbf{n}_l$,$\hat{\mathbf{I}}_2(\theta, \varphi, \gamma) = \hat{\mathbf{M}}\mathbf{n}_s$, (33) can be re-expressed as (76), shown at the bottom of the page. Here, $\hat{\mathbf{I}}_i(j)$ and $\tilde{\mathbf{I}}_i(j)$ are the jth element of $\hat{\mathbf{I}}_i(\theta, \varphi, \gamma)$ and $\tilde{\mathbf{I}}_i$, respectively, $j = 1, 2$, and $\hat{\mathbf{I}}_i(j)$ can be expressed as

$$\begin{cases} \hat{\mathbf{I}}_1(j) = \mathbf{E}_j^T\hat{\mathbf{I}}_1(\theta, \varphi, \gamma) = \mathbf{E}_j^T\mathbf{MR}(\theta, \varphi, \gamma)\mathbf{n}_l \\ \hat{\mathbf{I}}_2(j) = \mathbf{E}_j^T\hat{\mathbf{I}}_2(\theta, \varphi, \gamma) = \mathbf{E}_j^T\mathbf{MR}(\theta, \varphi, \gamma)\mathbf{n}_s \end{cases} \tag{88}$$

$$J_3(\Theta) = \arg\min_{\Theta}\left\{ \begin{array}{c} \sqrt{\left(2x_0\hat{\mathbf{I}}_1(1) - L_1\tilde{\mathbf{I}}_1(1)\right)^2 + \left(2x_0\hat{\mathbf{I}}_1(2) - L_1\tilde{\mathbf{I}}_1(2)\right)^2} \\ +\sqrt{\left(2y_0\hat{\mathbf{I}}_2(1) - L_2\tilde{\mathbf{I}}_2(1)\right)^2 + \left(2y_0\hat{\mathbf{I}}_2(2) - L_2\tilde{\mathbf{I}}_2(2)\right)^2} \end{array} \right\} \tag{76}$$

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where $\mathbf{E}_1 = [\,1\ 0\,]^T$, $\mathbf{E}_2 = [\,0\ 1\,]^T$. Then the derivation of the five parameters Θ can be expressed as

$$\begin{cases} \dfrac{\partial J_3(\Theta)}{\partial \Phi} = \dfrac{\sum_{j=1}^{2}\left(2x_0\hat{\mathbf{l}}_1(j) - L_1\tilde{\mathbf{l}}_1(j)\right)\cdot 2x_0\frac{\partial \hat{\mathbf{l}}_1(j)}{\partial \Phi}}{\sqrt{\sum_{j=1}^{2}\left(2x_0\hat{\mathbf{l}}_1(j) - L_1\tilde{\mathbf{l}}_1(j)\right)^2}} \\ \qquad\qquad + \dfrac{\sum_{j=1}^{2}\left(2y_0\hat{\mathbf{l}}_2(j) - L_2\tilde{\mathbf{l}}_2(j)\right)\cdot 2y_0\frac{\partial \hat{\mathbf{l}}_2(j)}{\partial \Phi}}{\sqrt{\sum_{j=1}^{2}\left(2y_0\hat{\mathbf{l}}_2(j) - L_2\tilde{\mathbf{l}}_2(j)\right)^2}} \\ \dfrac{\partial J_3(\Theta)}{\partial x_0} = \dfrac{\sum_{j=1}^{2}\left(2x_0\hat{\mathbf{l}}_1(j) - L_1\tilde{\mathbf{l}}_1(j)\right)\cdot 2\hat{\mathbf{l}}_1(j)}{\sqrt{\sum_{j=1}^{2}\left(2x_0\hat{\mathbf{l}}_1(j) - L_1\tilde{\mathbf{l}}_1(j)\right)^2}} \\ \dfrac{\partial J_3(\Theta)}{\partial y_0} = \dfrac{\sum_{j=1}^{2}\left(2y_0\hat{\mathbf{l}}_2(j) - L_2\tilde{\mathbf{l}}_2(j)\right)\cdot 2\hat{\mathbf{l}}_2(j)}{\sqrt{\sum_{j=1}^{2}\left(2y_0\hat{\mathbf{l}}_2(j) - L_2\tilde{\mathbf{l}}_2(j)\right)^2}} \end{cases} \tag{89}$$

where $(\partial\hat{\mathbf{l}}_i(j)/\partial\Phi)$ can be expressed as follows:

$$\begin{cases} \dfrac{\partial \hat{\mathbf{l}}_1(j)}{\partial \Phi} = \mathbf{E}_j^T\mathbf{M}\dfrac{\partial \mathbf{R}(\theta,\varphi,\gamma)}{\partial \Phi}\mathbf{n}_l \\ \dfrac{\partial \hat{\mathbf{l}}_2(j)}{\partial \Phi} = \mathbf{E}_j^T\mathbf{M}\dfrac{\partial \mathbf{R}(\theta,\varphi,\gamma)}{\partial \Phi}\mathbf{n}_s. \end{cases} \tag{90}$$

In summary, after obtaining the derivatives of $J_1(\theta,\varphi,\gamma)$, $J_2(\Theta)$ and $J_3(\Theta)$ for the five parameters, the derivatives of each parameter of $J(\Theta)$ can be obtained as follows:

$$\frac{\partial J(\Theta)}{\partial \Theta} = \frac{\partial J_1(\Theta)}{\partial \Theta} + \frac{\partial J_2(\Theta)}{\partial \Theta} + \frac{\partial J_3(\Theta)}{\partial \Theta}. \tag{91}$$

The gradient of $J(\Theta)$ can be obtained by bringing it into (59).

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